Chapter Eighteen

FRAUNHOFER DIFFRACTION I

No one has ever been able to define the difference between interference and diffraction satisfactorily. It is just a question of usage, and there is no specific, important physical difference between them. The best we can do is, roughly speaking, is to say that when there are only a few sources, say two, interfering, then the result is usually called interference, but if there is a large number of them, it seems that the word diffraction is more often used.

—Richard Feynman in *Feynman Lectures on Physics*, Vol. 1

Learning Objectives

After reading this chapter, the reader should be able to:

LO 1: describe diffraction phenomenon.
LO 2: derive the intensity distribution associated with the single-slit diffraction pattern.
LO 3: analyse diffraction by circular aperture.
LO 4: discuss the directionality of laser beams.
LO 5: discuss the limit of resolution.
LO 6: derive the intensity distributions associated with the 2 slit and N-slit Fraunhofer diffraction patterns.
LO 7: discuss the grating spectrun.
LO 8: discuss the diffraction of a plane wave incident obliquely on a grating.
LO 9: analyse X-ray diffraction and its experimental methods.
LO 10: discuss self-focussing or laser beams.

Important Milestones

1819 *Joseph Fraunhofer demonstrated the diffraction of light by gratings which were initially made by winding fine wires around parallel screws.*
1823 *Fraunhofer published his theory of diffraction.*
1835 *George Airy calculated the (Fraunhofer) diffraction pattern produced by a circular aperture.*

18.1 INTRODUCTION LO 1

Consider a plane wave incident on a long narrow slit of width b (see Fig. 18.1). According to geometrical optics one expects the region AB of the screen SS' to be illuminated and the remaining portion (known as the *geometrical shadow*) to be absolutely dark. However, if the observations are made carefully then one finds that if the width of the slit is not very large compared to the wavelength, then the light intensity in the region AB is not uniform and there is also some intensity inside the geometrical shadow. Further, if the width of the slit is made smaller, larger amounts of energy reach the geometrical shadow. This spreadingout of a wave when it passes through a narrow opening is usually referred to as diffraction and the intensity distribution on the screen is known as the **diffraction pattern**. We will discuss the

phenomenon of diffraction in this chapter and will show that the spreading out decreases with decrease in wavelength. Indeed, since the light wavelengths are very small ($\lambda \sim 5 \times 10^{-5}$ cm), the effects due to diffraction are not readily observed.



Fig. 18.1 If a plane wave is incident on an aperture then according to geometrical optics a sharp shadow will be cast in the region AB of the screen.

We should point out that there is not much of a difference between the phenomena of interference and diffraction, indeed, interference corresponds to the situation when we consider the superposition of waves coming out from a number of point sources and diffraction corresponds to the situation when we consider waves coming out from an area source like a circular or rectangular aperture or even a large number of rectangular apertures (like the diffraction grating).

The diffraction phenomena are usually divided into two categories: (i) Fresnel diffraction and (ii) Fraunhofer diffraction.

In the Fresnel class of diffraction, the source of light and the screen are, in general, at a finite distance from the diffracting aperture [see Fig. 18.2(a)]. In the Fraunhofer class of diffraction, the source and the screen are at infinite distances from the aperture; this is easily achieved by placing the source on the focal plane of a convex lens and placing the screen on the focal plane of another convex lens [see Fig. 18.2(b)]. The two lenses effectively moved the source and the screen to infinity because the first lens makes the light beam parallel and the second lens effectively makes the screen receive a parallel beam of light. It turns out that it is much easier to calculate the intensity distribution of a Fraunhofer diffraction pattern which we plan to do in this chapter. Further, the Fraunhofer diffraction pattern is not difficult to observe; all that one needs is an ordinary laboratory spectrometer; the collimator renders a parallel beam of light and the telescope receives parallel beams of light on its focal plane. The diffracting aperture is placed on the prism table. In Chapter 20, we will study the Fresnel class of diffraction and will discuss the transition from the Fresnel region to the Fraunhofer region.

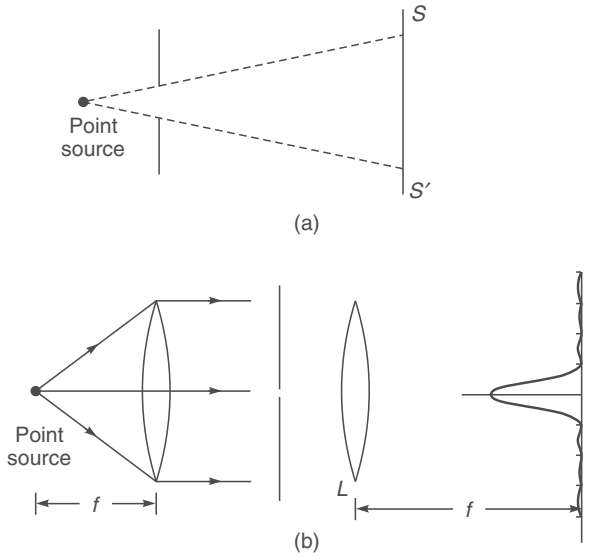


Fig. 18.2 (a) When either the source or the screen (or both) are at finite distances from the aperture, the diffraction pattern corresponds to the Fresnel class. (b) In the Fraunhofer class both the source and the screen are at infinity.

18.2 SINGLE-SLIT DIFFRACTION PATTERN

LO 2

We will first study the Fraunhofer diffraction pattern produced by an infinitely long slit of width b. A plane wave is assumed to fall normally on the slit and we wish to calculate the intensity distribution on the focal plane of the lens L [see Fig. 18.3(a)]. We assume that the slit consists of a large number of equally spaced point sources and that each point on the slit is a source of Huygens' secondary wavelets which interfere with the wavelets emanating from other points. Let the point sources be at $A_1, A_2, A_3, \ldots$ and let the distance between two consecutive points be Δ [see Fig. 18.3(b)]. Thus, if the number of point sources be n, then

$$b = (n-1)\Delta \qquad \textbf{(18.1)}$$

We will now calculate the resultant field produced by these n sources at the point P, P being an arbitrary point (on the focal plane of the lens) receiving parallel rays making an angle θ with the normal to the slit [see Fig. 18.3(b)]. Since the slit actually consists of a continuous distribution of sources, we will, in the final expression, let n go to infinity and Δ go to zero such that $n\Delta$ tends to b.

Now, at the point P, the amplitudes of the disturbances reaching from $A_1, A_2, \ldots$ will be very nearly the same because the point P is at a distance which is very large in comparison to b [see Fig. 18.3(b)]. However, because of even slightly different path lengths to the point P, the field produced by A_1 will differ in phase from the field produced by A_2.

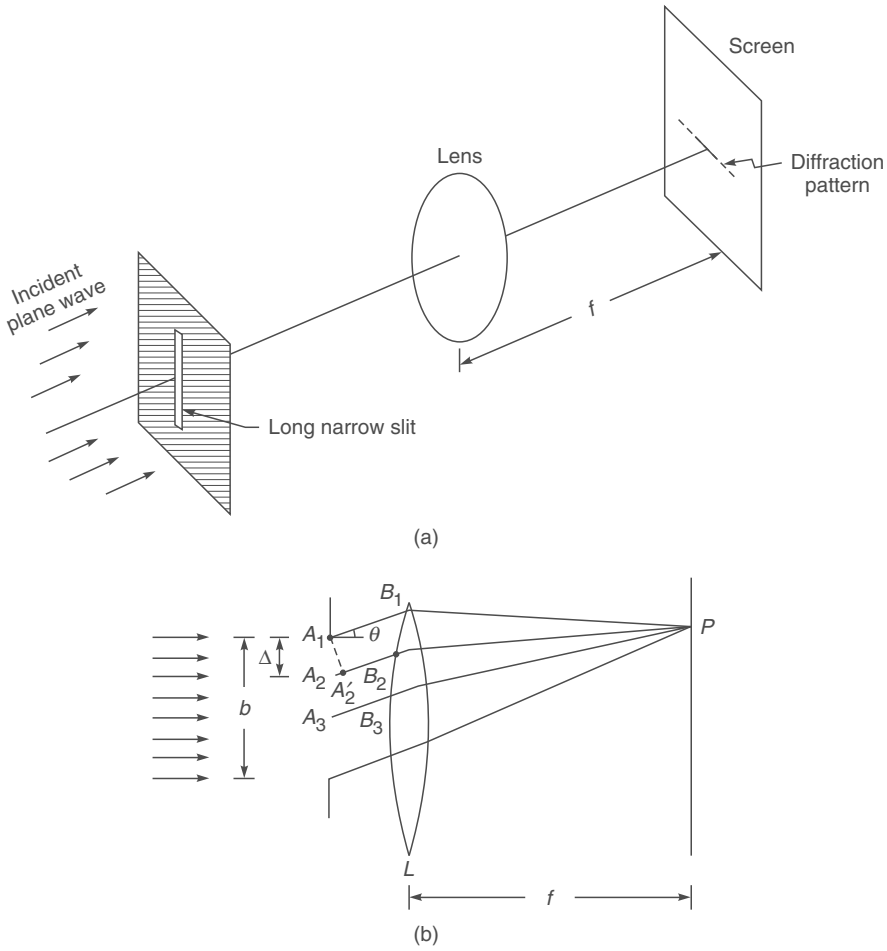


Fig. 18.3 (a) Diffraction of a plane wave incident normally on a long narrow slit of width b. Notice that the spreading occurs along the width of the slit. (b) In order to calculate the diffraction pattern, the slit is assumed to consist of a large number of equally spaced points.

For an incident plane wave, the points A_1, A_2,... are in phase and, therefore, the additional path traversed by the disturbance emanating from the point A_2 will be A_2A_2' where A_2' is the foot of the perpendicular drawn from A_1 on A_2B_2. This follows from the fact that the optical paths A_1B_1P and $A_2'B_2P$ are the same. If the diffracted rays make an angle θ with the normal to the slit then the path difference would be

$$A_2A_2' = \Delta \sin \theta$$

The corresponding phase difference, ϕ, would be given by

$$\phi = \frac{2\pi}{\lambda} \Delta \sin \theta \qquad \textbf{(18.2)}$$

Thus, if the field at the point P due to the disturbance emanating from the point A_1 is $a \cos \omega t$ then the field due to the disturbance emanating from A_2 would be $a \cos (\omega t - \phi)$. Now the difference in the phases of the disturbance reaching from the points A_2 and A_3 will also be ϕ and thus the resultant field at the point P would be given by

$$E = a[\cos \omega t + \cos (\omega t - \phi) + \ldots + \cos (\omega t - (n-1)\phi)] \qquad \textbf{(18.3)}$$

where

$$\phi = \frac{2\pi}{\lambda} \Delta \sin \theta$$

Now, we had shown in Sec. 11.7 that

$$\cos \omega t + \cos (\omega t - \phi) + \ldots + \cos [\omega t - (n-1)\phi]$$
$$= \frac{\sin n\phi/2}{\sin \phi/2} \cos\left[\omega t - \frac{1}{2}(n-1)\phi\right] \qquad \textbf{(18.4)}$$

Thus,

$$E = E_0 \cos\left[\omega t - \frac{1}{2}(n-1)\phi\right] \qquad \textbf{(18.5)}$$

where the amplitude E_θ of the resultant field would be given by*

$$E_\theta = a\frac{\sin(n\phi/2)}{\sin\phi/2} \tag{18.6}$$

In the limit of $n \to \infty$ and $\Delta \to 0$ in such a way that $n\Delta \to b$, we have

$$\frac{n\phi}{2} = \frac{\pi}{\lambda} n\Delta \sin\theta \to \frac{\pi}{\lambda} b \sin\theta$$

Further,

$$\phi = \frac{2\pi}{\lambda}\Delta \sin\theta = \frac{2\pi}{\lambda}\frac{b\sin\theta}{n}$$

would tend to zero and we may, therefore, write

$$E_\theta \approx \frac{a\sin\left(\frac{n\phi}{2}\right)}{\frac{\phi}{2}} = na\,\frac{\sin\frac{\pi b\sin\theta}{\lambda}}{\frac{\pi b\sin\theta}{\lambda}}$$

$$= A\,\frac{\sin\beta}{\beta} \tag{18.7}$$

where**

$$A = na$$

and

$$\beta = \frac{\pi b\sin\theta}{\lambda} \tag{18.8}$$

Thus,

$$E = A\,\frac{\sin\beta}{\beta}\cos(\omega t - \beta) \tag{18.9}$$

The corresponding intensity distribution is given by

$$I = I_0\,\frac{\sin^2\beta}{\beta^2} \tag{18.10}$$

where I_0 represents the intensity at $\theta = 0$

18.2.1 Positions of Maxima and Minima

The variation of the intensity with β is shown in Fig. 18.4(a). It is obvious from Eq. (18.10) that the intensity is zero when

$$\beta = m\pi,\ m \neq 0 \tag{18.11}$$

[When $\beta = 0$, $\frac{\sin\beta}{\beta} = 1$ and $I = I_0$ which corresponds to the maximum of the intensity.] Substituting the value of β one obtains

$$b\sin\theta = m\lambda;\ m = \pm 1, \pm 2, \pm 3, \ldots \text{ (minima)} \tag{18.12}$$

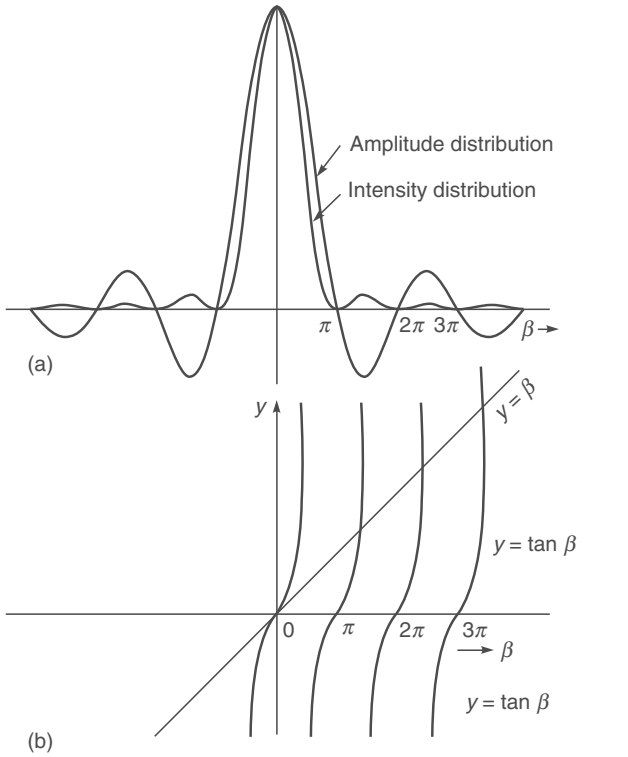


Fig. 18.4 (a) The intensity distribution corresponding to the single slit Fraunhofer diffraction pattern. (b) Graphical method for determining the roots of the equation $\tan\beta = \beta$.

as the conditions for minima. The first minimum occurs at $\theta = \pm\sin^{-1}\left(\frac{\lambda}{b}\right)$; the second minimum at $\theta = \pm\sin^{-1}\left(\frac{2\lambda}{b}\right)$, etc. Since $\sin\theta$ cannot exceed unity, the maximum value of m is the integer which is less than (and closest to) $\frac{b}{\lambda}$.

The positions of minima can directly be obtained by simple qualitative arguments. Let us consider the case $m = 1$. The angle θ satisfies the equation

$$b\sin\theta = \lambda \tag{18.13}$$

We divide the slit into two halves as shown in Fig. 18.5. Consider two points A and A' separated by a distance $b/2$. Clearly the path difference between the disturbances (reaching the point P) emanating from A and A' is $\frac{b}{2}\sin\theta$ which in this case is $\frac{\lambda}{2}$. The corresponding phase difference will be π and the resultant disturbance will be zero. Similarly, the disturbance from the point B will be cancelled by the disturbance reaching from the point B'. Thus, the resultant

* Equation (18.6) represents the amplitude distribution due to the interference of n point sources. Thus, for $n = 2$, the amplitude E_θ becomes $\cos\phi/2$ giving rise to $\cos^2\phi/2$ intensity distribution [cf. Eq. (14.13) of Chapter 14]. Notice that if we have a large number of equidistant sources oscillating in phase, then the propagation is only in cetain directions where the displacements add up in phase.

** We may mention here that in the limit $n \to \infty$ and $a \to 0$ the product na tends to a finite limit.

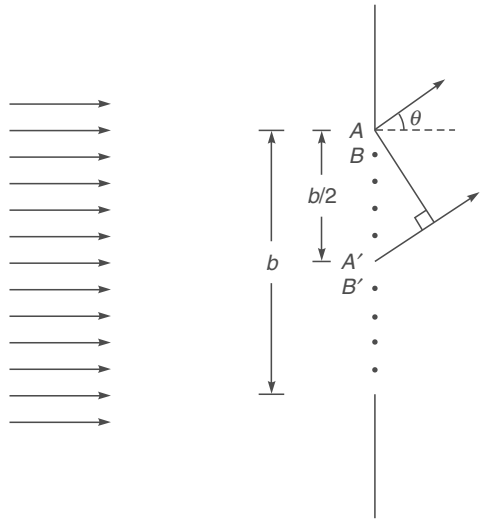


Fig. 18.5 The slit is divided into two halves for deriving the condition for the first minimum.

disturbance due to the upper half of the slit will be canceled by the disturbances reaching from the lower half and the resultant intensity will be zero. In a similar manner when

$$b \sin \theta = 2\lambda \qquad \textbf{(18.14)}$$

we divide the slit into four parts; the first and second quarters cancelling each other and the third and fourth quarters cancelling each other. Similarly when $m = 3$, the slit is divided into six parts and so on.

In order to determine the positions of maxima, we differentiate Eq. (18.10) with respect to β and set it equal to zero. Thus

$$\frac{dI}{d\beta} = I_0 \left[\frac{2 \sin \beta \cos \beta}{\beta^2} - \frac{2 \sin^2 \beta}{\beta^3} \right] = 0$$

or

$$\sin \beta [\beta - \tan \beta] = 0 \qquad \textbf{(18.15)}$$

The condition $\sin \beta = 0$, or $\beta = m\pi$ ($m \neq 0$) correspond to minima. The conditions for maxima are roots of the following transcendental equation

$$\tan \beta = \beta \quad \text{(maxima)} \qquad \textbf{(18.16)}$$

The root $\beta = 0$ corresponds to the central maximum. The other roots can be found by determining the points of intersections of the curves $y = \beta$ and $y = \tan \beta$ [see Fig. 18.4(b)]. The intersections occur at $\beta = 1.43\pi$, $\beta = 2.46\pi$, etc., and are known as the first maximum, the second maximum, etc. Since

$$\left[\frac{\sin (1.43\pi)}{1.43\pi} \right]^2$$

is about 0.0496, the intensity of the first maximum is about 4.96% of the central maximum. Similarly, the intensities of the second and third maxima are about 1.68% and 0.83% of the central maximum, respectively.

Example 18.1 A parallel beam of light is incident normally on a narrow slit of width 0.2 mm. The Fraunhofer diffraction pattern is observed on a screen which is placed at the focal plane of a convex lens whose focal length is 20 cm. Calculate the distance between the first two minima and the first two maxima on the screen. Assume $\lambda = 5 \times 10^{-5}$ cm and that the lens is placed very close to the slit.

Solution:

$$\frac{\lambda}{b} = \frac{5 \times 10^{-5}}{2 \times 10^{-2}} = 2.5 \times 10^{-3}$$

Now, the conditions for diffraction minima are given by $\sin \theta = m\lambda/b$. We assume θ to be small (measured in radians) so that we may write $\sin \theta \approx \theta$ (an assumption which will be justified by subsequent calculations); thus, on substituting the value of λ/b, we get

$$\theta \simeq 2.5 \times 10^{-3} \text{ and } 5 \times 10^{-3} \text{ radians}$$

as the angles of diffraction corresponding to the first and second minima, respectively. Notice that since

$$\sin (2.5 \times 10^{-3}) = 2.4999973 \times 10^{-3}$$

the error in the approximations $\sin \theta \simeq \theta$ is about 1 part in a million! These minima will be separated by a distance $(5 \times 10^{-3} - 2.5 \times 10^{-3}) \times 20 = 0.05$ cm on the focal plane of the lens. Similarly, the first and second maxima occur at

$$\beta = 1.43\pi \quad \text{and} \quad 2.46\pi$$

respectively. Thus,

$$b \sin \theta = 1.43\lambda \quad \text{and} \quad 2.46\lambda$$

or

$$\sin \theta = 1.43 \times 2.5 \times 10^{-3} \quad \text{and} \quad 2.46 \times 2.5 \times 10^{-3}$$

Consequently, the maxima will be separated by the distance given by

$$(2.46 - 1.43) \times 2.5 \times 10^{-3} \times 20 \simeq 0.05 \text{ cm}$$

Example 18.2 Consider, once again, a parallel beam of light ($\lambda = 5 \times 10^{-5}$ cm) to be incident normally on a long narrow slit of width 0.2 mm. A screen is placed at a distance of 3 m from the slit. Assuming that the screen is so far away that the diffraction is essentially of the Fraunhofer type, calculate total width of the central maximum.

Solution: As in Example 18.1, the first minimum occurs at $\theta \simeq 2.5 \times 10^{-3}$ radians; thus the total width of the central maximum is approximately given by

$$2 \times 3 \times \tan (2.5 \times 10^{-3}) \simeq 0.015 \text{ m}$$

In Fig. 18.6, we have given the actual single slit diffraction pattern (as seen on a screen) for the following values of slit widths: 8.8×10^{-3} cm, 1.76×10^{-2} cm, 3.5×10^{-2} cm and 7.0×10^{-2} cm. The light wavelength used was 6328 Å = 6.328×10^{-5} cm. We may note the following two points:

(i) The spreading is only in the direction of the width of the slit. This is because of the fact that the lengths of the slits were very large compared to their widths.

(ii) The values of λ/b corresponding to the four slit widths are 7.191×10^{-3}, 3.595×10^{-3}, 1.808×10^{-3} and 0.904×10^{-3}. Thus, the diffraction angle at which the first minimum will occur will be

$$\theta \simeq \sin \theta = 7.191 \times 10^{-3}, 3.595 \times 10^{-3}, 1.808 \times 10^{-3} \text{ and } 0.904 \times 10^{-3};$$

where the angles are measured in radians.* The intensity distributions predicted by Eq. (18.10) are given in Fig. 18.7 for $b = 8.8 \times 10^{-3}$ cm and 1.76×10^{-2} cm. For $b >> \lambda$, most of the energy (of the diffracted beam) is contained between the first two minima, i.e., for

$$-\frac{\lambda}{b} \lesssim \theta \lesssim \frac{\lambda}{b} \qquad \textbf{(18.17)}$$

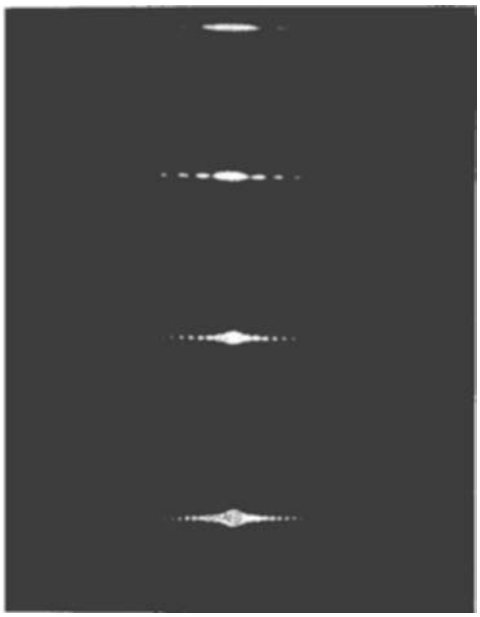


Fig. 18.6 The single-slit diffraction patterns corresponding to b = 0.0088, 0.0176, 0.035 and 0.070 cm, respectively. The wavelength of the light used is 6.328×10^{-5} cm (Adapted from Ref. 18.17; used with permission).

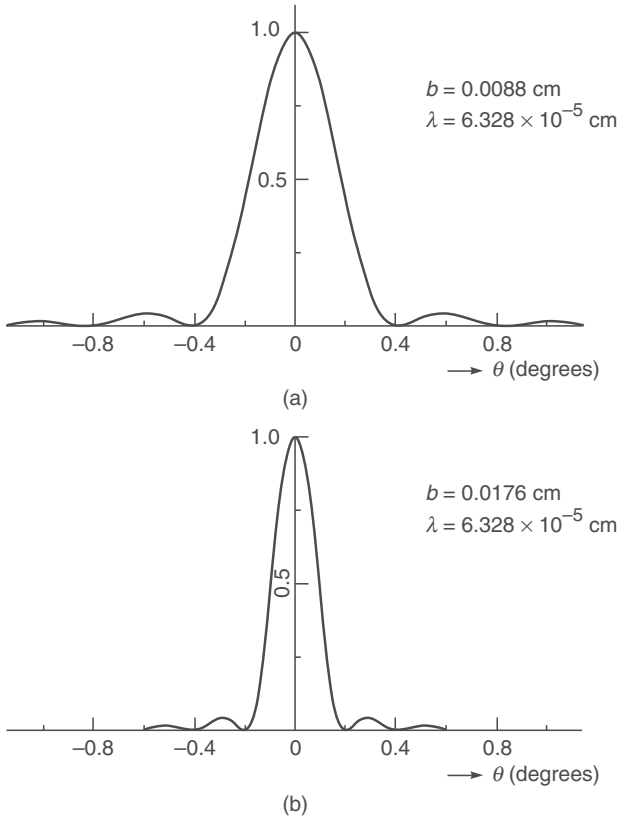


Fig. 18.7 The intensity distribution as calculated by using Eq. (18.10) for b = 0.0088 cm and 0.0176 cm ($\lambda = 6.328 \times 10^{-5}$ cm).

(where θ is measured in radians). Thus the divergence angle (which would contain most of the energy) would be given by

$$\Delta\theta \sim \frac{\lambda}{b} \qquad \textbf{(18.18)}$$

For very small values of b, the light almost uniformly spreads out from the slit. We should also mention that in the limit of $\lambda \to 0$, $\Delta\theta \to 0$ and the diffraction effects are absent.

18.3 DIFFRACTION BY A CIRCULAR APERTURE LO 3

In the previous section, we have shown that when a plane wave is incident on a long narrow slit (of width b), then the emergent wave spreads out (along the width of the slit) with angular divergence $\sim \lambda/b$. In a similar manner, one can discuss the diffraction of a plane wave by a circular aperture. Figure 18.8 shows the arrangement for observing the

* Figure 18.6 corresponds to the photographic film being 15 feet away from the slit. Thus it records the Fraunhofer pattern (see also Sec. 20.7) and for $b = 8.8 \times 10^{-3}$ cm, 1.76×10^{-2} cm, 3.5×10^{-2} cm and 7.0×10^{-2} cm, the first minima occur at distances of 3.288 cm, 1.644 cm, 0.827 cm and 0.413 cm, respectively, from the central maximum.

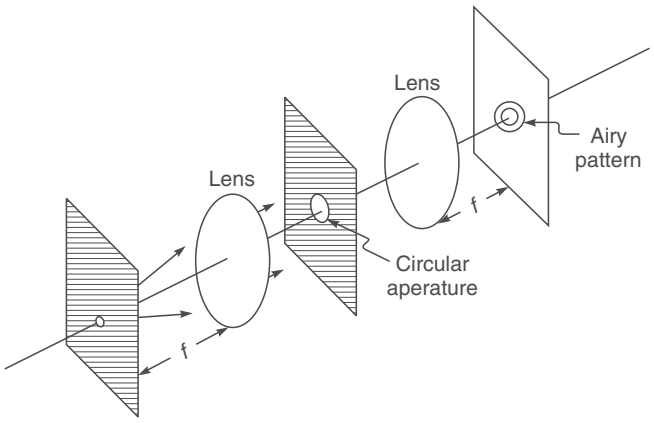


Fig. 18.8 Experimental arrangement for observing the Fraunhofer diffraction pattern by a circular aperture.

diffraction pattern; a plane wave is incident normally on the circular aperture and a lens whose diameter is much larger than that of the aperture is placed close to the aperture and the Fraunhofer diffraction pattern is observed on the focal plane of the lens. Because of the rotational symmetry of the system, the diffraction pattern will consist of concentric dark and bright rings; this diffraction pattern (as observed on the back focal plane of the lens) is known as the **Airy pattern**. In Figs. 18.9(a) and (b), we have shown the Airy patterns corresponding to the radius of the circular aperture being 0.5 mm and 0.25 mm, respectively. The detailed derivation of the diffraction pattern for a circular aperture is somewhat complicated (see Sec. 19.7); we give here the final result: the intensity distribution is given by

$$I = I_0 \left[\frac{2J_1(v)}{v}\right]^2 \qquad \textbf{(18.19)}$$

where

$$v = \frac{2\pi}{\lambda}\, a \sin\theta \qquad \textbf{(18.20)}$$

a being the radius of the circular aperture, λ the wavelength of light and θ the angle of diffraction; I_0 is the intensity at $\theta = 0$ (which represents the central maximum) and $J_1(v)$ is known as the **Bessel function of the first order.** On the focal plane of the convex lens

$$v \approx \frac{2\pi}{\lambda}\, a \frac{(x^2 + y^2)^{\frac{1}{2}}}{f} \qquad \textbf{(18.21)}$$

where f is the focal length of the lens. For those not familiar with Bessel functions, we may mention that the variation of $J_1(v)$ is somewhat like a damped sine curve (see Fig. 18.10) and although $J_1(0) = 0$, we have

$$\underset{v \to 0}{Lt} \frac{2\,J_1(v)}{v} = 1$$

similar to the relation

$$\underset{x \to 0}{Lt} \frac{\sin x}{x} = 1$$

Other zeros of $J_1(v)$ occur at

$$v = 3.832,\ 7.016,\ 10.174,\ \ldots$$

In Fig. 18.11, we have plotted the function

$$\left[\frac{2\,J_1(v)}{v}\right]^2$$

which represents the intensity distribution corresponding to the Airy pattern. Thus the successive dark rings in the Airy pattern (see Fig. 18.9) will correspond to

$$v = \frac{2\pi}{\lambda}\, a \sin\theta$$

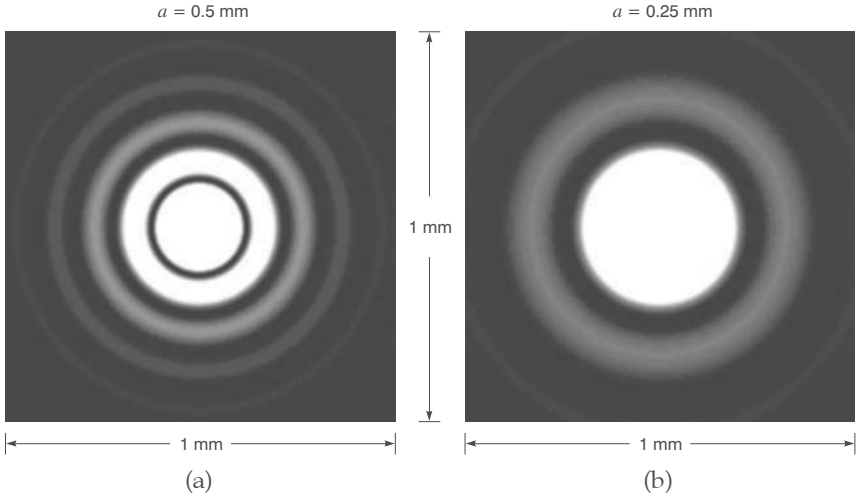


Fig. 18.9 Computer generated Airy patterns; (a) and (b) correspond to a = 0.5 mm and a = 0.25 mm, respectively at the focal plane of a lens of focal length 20 cm (λ = 0.5 μm).

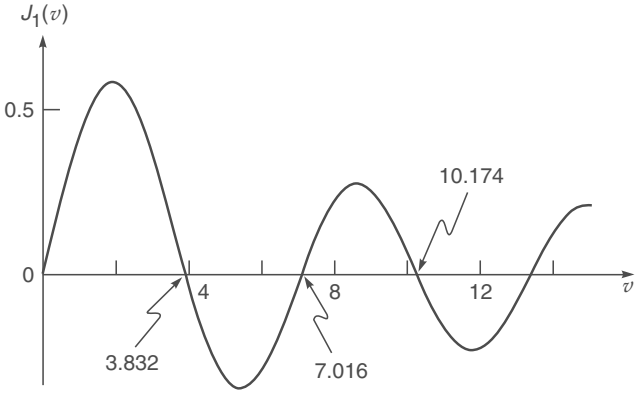


Fig. 18.10 The variation of $J_1(v)$ with v.

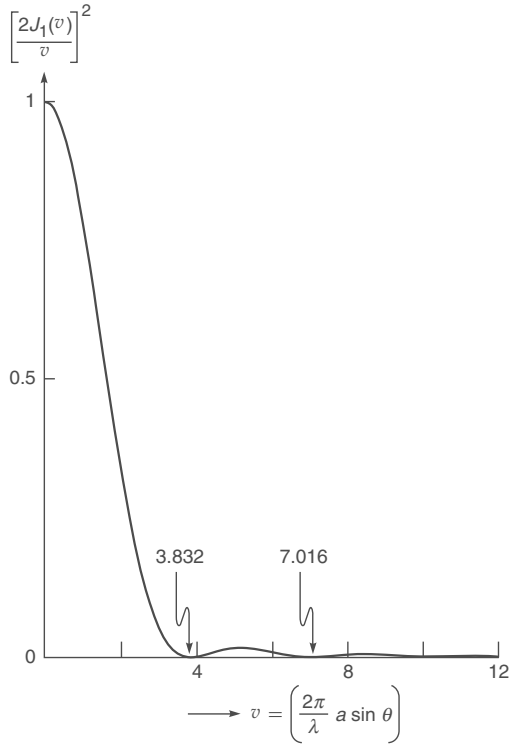


Fig. 18.11 The intensity variation associated with the Airy pattern.

$$= 3.832, 7.016, 10.174, \ldots \quad \textbf{(18.22)}$$

or

$$\sin\theta = \frac{3.832\,\lambda}{2\pi a}, \frac{7.016\,\lambda}{2\pi a}, \ldots \quad \textbf{(18.23)}$$

If f represents the focal length of the convex lens, then the Radii of the dark rings

$$= f\tan\theta \approx \frac{3.832\,\lambda\, f}{2\pi a}, \frac{7.016\,\lambda\, f}{2\pi a}, \ldots \quad \textbf{(18.24)}$$

where we have assumed θ to be small so that $\tan\theta \approx \sin\theta$. The Airy patterns shown in Figs. 18.9(a) and (b) correspond to $a = 0.5$ mm and 0.25 mm, respectively; both figures correspond to $\lambda = 5000$Å and $f = 20$ cm. Thus

Radius of the first dark ring ≈ 0.12 mm and 0.24 mm

corresponding to $a = 0.5$ mm and 0.25 mm, respectively. Detailed mathematical analysis shows that about 84% of the energy is contained within the first dark ring (see Sec. 19.7); thus we may say that the angular spread of the beam is approximately given by

$$\Delta\theta \approx \frac{0.61\,\lambda}{D} \approx \frac{\lambda}{D} \quad \textbf{(18.25)}$$

where D (= $2a$) represents the diameter of the aperture. Comparing Eqs. (18.18) and (18.25), we may say that the angular divergence associated with the diffraction pattern can be written in the following general form:

$$\Delta\theta \sim \frac{\lambda}{\text{Linear dimension of the aperture}} \quad \textbf{(18.26)}$$

In a dark room, if we make a laser beam incident normally on a pin-hole, we will be able to see the Airy Pattern on the wall.

Example 18.3 Calculate the radii of the first two dark rings of the Fraunhofer diffraction pattern produced by a circular aperture of radius 0.02 cm at the focal plane of a convex lens of focal length 20 cm. Assume $\lambda = 6 \times 10^{-5}$ cm.

Solution: The first dark ring occurs at

$$\theta \approx \sin\theta = \frac{1.22 \times 6 \times 10^{-5}}{2 \times 0.02} \approx 1.8 \times 10^{-3} \text{ radians}$$

Thus, the radius of the first dark ring is

$$\approx 20 \times 1.8 \times 10^{-3} = 3.6 \times 10^{-2} \text{ cm}$$

Similarly, the radius of the second dark ring is

$$\approx 20 \times \frac{7.016 \times 6 \times 10^{-5}}{2\pi \times 0.02} \approx 6.7 \times 10^{-2} \text{ cm}$$

In Fig. 18.12, we have shown that if an obstacle with a small gap is placed in the tank the ripples emerge in an almost semicircular pattern; the small gap acting almost like a point source. If the gap is large, the diffraction is much more limited. *Small*, in this context, means that the size of the obstacle is comparable to the wavelength of the ripples.

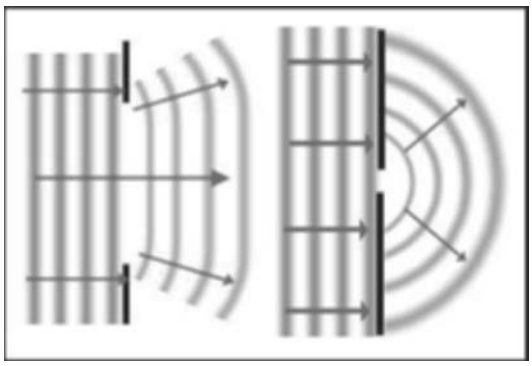


Fig. 18.12 If an obstacle with a small gap is placed in the tank the ripples emerge in an almost semicircular pattern; the small gap acting almost like a point source. If the gap is large however, the diffraction is much more limited. *Small*, in this context, means that the size of the obstacle is comparable to the wavelength of the ripples. The drawing is by Ms. Theresa Knott; used with her kind permission. [*A colored diagram appears as Fig. 22 in the prelim pages.*]

18.4 DIRECTIONALITY OF LASER BEAMS

LO 4

An ordinary source of light (like a sodium lamp) radiates in all directions. On the other hand, the divergence of a laser beam is primarily due to diffraction effects. For most laser beams, the transverse amplitude distribution is approximately Gaussian; indeed just when the beam is leaving the laser (which we assume to be $z = 0$), the amplitude distribution can be assumed to be given by

$$A(x,y) = a \exp\left[-\frac{x^2+y^2}{w_0^2}\right] \qquad \textbf{(18.27)}$$

where we have assumed that the phase front is plane at $z = 0$. From the above equation it follows that at a distance w_0 from the z-axis, the amplitude falls by a factor $1/e$ (i.e., the intensity reduces by a factor $1/e^2$). This quantity w_0 is called the *spot size* of the beam. In Sec. 20.5 (and Appendix D) we will show that as the beam propagates in the z-direction, the intensity distribution is given by

$$I(x,y,z) = \frac{I_0}{1+\gamma^2} \exp\left[-\frac{2(x^2+y^2)}{w^2(z)}\right] \qquad \textbf{(18.28)}$$

where

$$\gamma = \frac{\lambda z}{\pi w_0^2}$$

$$w(z) = w_0\sqrt{1+\gamma^2}$$

$$= w_0\left[1+\frac{\lambda^2 z^2}{\pi^2 w_0^4}\right]^{1/2} \qquad \textbf{(18.29)}$$

Thus the transverse intensity distribution remains Gaussian with the beam-width increasing with z. For large values of $z\left(>>\frac{w_0^2}{\lambda}\right)$, we obtain

$$w(z) \approx w_0 \frac{\lambda z}{\pi w_0^2} = \frac{\lambda z}{\pi w_0} \qquad \textbf{(18.30)}$$

which shows that the width increases linearly with z. We define the diffraction angle as

$$\tan\theta = \frac{w(z)}{z} \approx \frac{\lambda}{\pi w_0} \qquad \textbf{(18.31)}$$

showing that the rate of increase in the width is proportional to the wavelength and inversely proportional to the initial width of the beam; the above equation is consistent with Eq. (18.26).

We consider the propagation of a Gaussian beam with $\lambda = 0.5\ \mu$m. If the initial spot size is given by $w_0 = 1$ mm, then

$$2\theta \approx 0.018^\circ \text{ and at } z = 10 \text{ m, we get } w \approx 1.88 \text{ mm.}$$

[We must use Eq. (18.28) and not Eq. (18.31) — why??]. Similarly, for $w_0 = 0.25$ mm,

$$2\theta \approx 0.073^\circ \text{ and at } z = 10 \text{ m, we get } w \approx 6.35 \text{ mm.}$$

(see Fig. 18.13). Notice that θ increases with decrease in w_0 (smaller the size of the aperture, greater is the diffraction).

From Eq. (18.31) we find that

(a) For a given value of λ, θ increases with a decrease in the value of w_0; this implies that smaller the initial spot size of the beam greater will be the diffraction divergence.

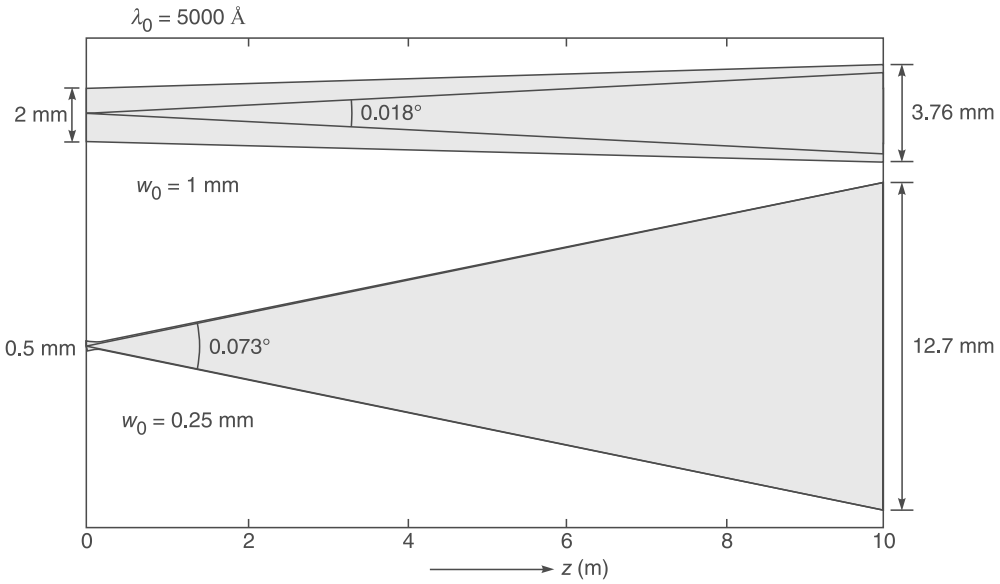


Fig. 18.13 Diffraction divergence of a Gaussian beam whose phase front is plane at $z = 0$. The figure shows the increase in the diffraction divergence as the initial spot size is decreased from 1 mm to 0.25 mm; the wavelength is assumed to be 5000 Å.

(b) For a given value of w_0, the value of θ (and hence the diffraction divergence) decreases with decrease in the value of λ. In Fig. 18.14, we have shown decrease in diffraction divergence for $w_0 = 0.25$ mm as the wavelength is decreased from 5000 Å to 500 Å; indeed as $\lambda \to 0$, $\theta \to 0$ and there is no diffraction which is the geometric optics limit.

In Fig. 18.15, we have shown a laser beam propagating through the atmosphere. Notice the small divergence of the beam. From Eq. (18.27) one can readily show that

$$\int_{-\infty}^{+\infty}\int I(x, y, z)\, dx\, dy = \frac{\pi w_0^2}{2} I_0$$

which is independent of z. This is to be expected, as the total energy crossing the entire x-y plane will not change with z.

Example 18.4 The output from a single-mode fiber operating at the He–Ne laser wavelength ($\lambda_0 = 0.6328$ μm) is approximately Gaussian with $w_0 = 5$ μm. Thus, the corresponding divergence is

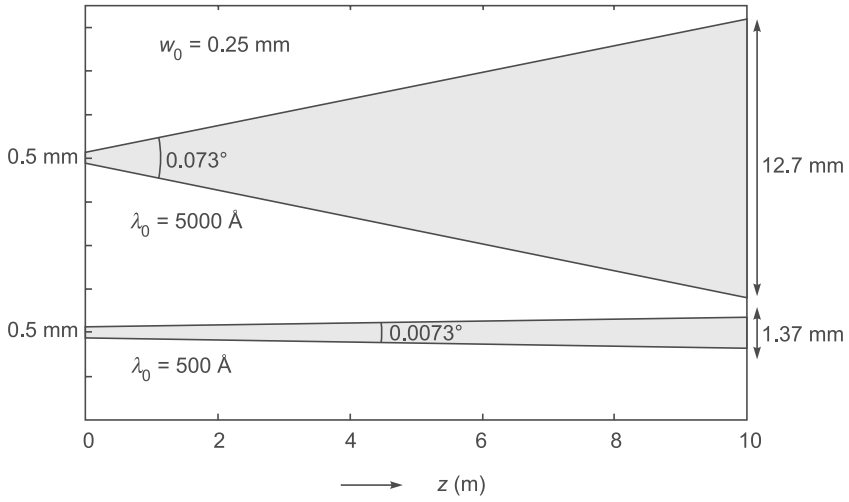


Fig. 18.14 Diffraction divergence of a Gaussian beam whose phase front is plane at $z = 0$. The figure shows the decrease in divergence as the wavelength is decreased from 5000 Å to 5000 Å; the initial spot size (w_0) is assumed to be 0.25 mm.

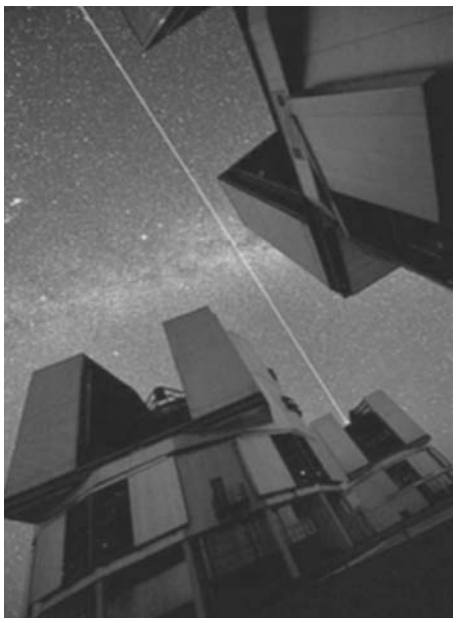


Fig. 18.15 The laser beam, launched from VLT's 8.2-metre Yepun telescope, crosses the sky and creates an artificial star at 90 km altitude in the high Earth's mesosphere. Notice that the spreading of the beam is extremely small. [*Photograph courtesy: Dr. G Huedepohl. The photograph is in the public domain. A color photograph appears as Fig. 23 on the prelim pages of the book*].

$$\theta \approx \tan^{-1}\left(\frac{\lambda_0}{\pi w_0}\right) \approx 2.3^\circ$$

Thus, if a screen as placed at a distance of about 50 cm from the fiber, the radius of the beam is about 2 cm.

A beam is said to be *diffraction limited* if it diverges only due to diffraction. Usually laser beams are diffraction limited. On the other hand, if we have a tiny filament at the focal plane of a lens, the beam will diverge primarily due to the finite size of the filament (see Fig. 18.16). The angular spread of the beam is given by (see Fig. 18.16)

$$\Delta\theta \approx \frac{l}{f} \tag{18.32}$$

where l is the length of the filament and f the focal length of the lens. If the linear dimension of the filament is about 2 mm (placed on the focal plane of a convex lens of focal length 10 cm) then the angular divergence of the beam (due to the finite size of the filament) is approximately given by

$$\Delta\theta \approx \frac{2\text{ mm}}{100\text{ mm}} = 0.02 \text{ radians}$$

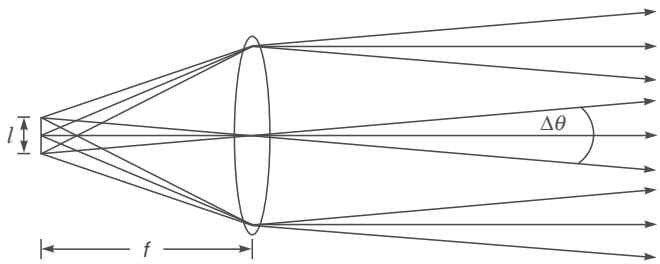


Fig. 18.16 A filament placed at the focal plane of a convex lens.

If the diameter of the aperture of the lens is 5 cm then the angular divergence due to diffraction would be

$$\Delta\theta \approx \frac{\lambda}{D} \approx \frac{5\times10^{-5}\text{ cm}}{5\text{ cm}} = 0.00001 \text{ radians}$$

which is much much smaller than the angular divergence of the beam due to the finite size of the filament. Only if the size of the filament is smaller than 10^{-3} mm would the beam divergence be determined by diffraction. Thus for most practical sources, the beam divergence is due to the finite size of the filament rather than by diffraction.

18.4.1 Focusing of Laser Beams

As mentioned earlier, laser beams are usually diffraction limited. If such a diffraction limited beam is allowed to fall on a convex lens then

$$\text{Radius of the focussed spot} \approx \frac{\lambda_0 f}{a} \tag{18.33}$$

(see Fig. 18.17). In Eq. (18.32), f represents the focal length of the lens and a represents the beam radius or the radius of the aperture of the lens (whichever is smaller). Thus,

$$\text{Area of the focussed spot } A \approx \pi\left(\frac{\lambda_0 f}{a}\right)^2$$

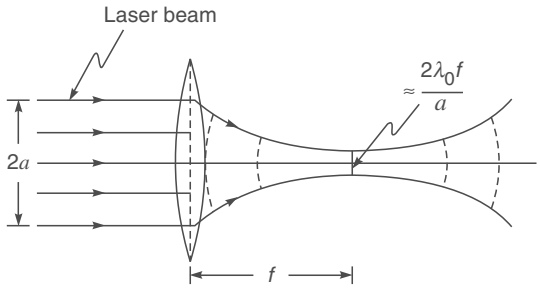


Fig. 18.17 If a truncated plane wave (of diameter $2a$) is incident on an aberrationless lens of focal length f, then the wave emerging from the lens will get focused to spot of radius $\approx \lambda_0 f/a$; the area of the focused spot size is $\approx \pi(\lambda_0 f/a)^2$.

We illustrate the effects of this focusing through some examples.

Example 18.5 We consider a 2 mW laser beam ($\lambda_0 \approx 6 \times 10^{-5}$ cm) incident on the eye whose focal length is given by $f \approx 2.5$ cm. If the pupil diameter ($= 2a$) is taken to be 2 mm, then

$$\text{Area of the focused spot } A = \pi \left(\frac{\lambda_0 f}{a} \right)^2 \approx 7 \times 10^{-6} \text{ cm}^2$$

On the retina, the intensity will be approximately given by

$$I \approx \frac{P}{A} \approx \frac{2 \times 10^{-3} \text{ W}}{7 \times 10^{-10} \text{ m}^2} \approx 3 \times 10^6 \text{ W/m}^2$$

Such high intensities will damage the retina!!!! **So never look into a (seemingly innocent) low power laser beam.**

Example 18.6 We next consider a 3 MW laser beam ($\lambda_0 \approx 6 \times 10^{-5}$ cm and beam width $2a \approx 1$ cm) incident on a lens of focal length of 5 cm, then

Area of the focused spot

$$A = \pi \left(\frac{\lambda_0 f}{a} \right)^2 \approx 10^{-6} \text{ cm}^2 = 10^{-10} \text{ m}^2$$

On the focal plane of the lens, the intensity will be approximately given by

$$I \approx \frac{P}{A} \approx \frac{3 \times 10^6 \text{ W}}{10^{-10} \text{ m}^2} \approx 3 \times 10^{16} \text{ W/m}^2$$

Now, the intensity of the beam is related to the electric field amplitude E_0 through the following relation [see Eq. (23.78)

$$I = \frac{1}{2} \varepsilon_0 c E_0^2 \qquad \textbf{(18.34)}$$

where $\varepsilon_0 \approx 8.854 \times 10^{-12}$ MKS units represents the dielectric permittivity of free space and $c \approx 3 \times 10^8$ m/s represents the speed of light in free space. Substituting $I \approx 3 \times 10^{16}$ W/m^2 in Eq. (18.34) we readily get

$$E_0 \approx 5 \times 10^9 \text{ V/m}$$

Such high electric fields results in the creation of spark in air (see Fig. 18.18). Thus laser beams (because of their high directionality) can be focused to extremely small regions producing very high intensities. Such high intensities lead to numerous industrial applications of the laser such as welding, hole drilling, cutting materials, etc (see e.g., Ref. 18.5). In Fig. 18.19, we have shown a focussed laser beam drilling through concrete.

In the following two examples, we will calculate the intensities (at the retina of our eye) when we directly view a 500 W bulb or the Sun **(Caution: Never look into the Sun; the retina will be damaged not only because of high intensities but also because of large ultraviolet content of the sunlight).**

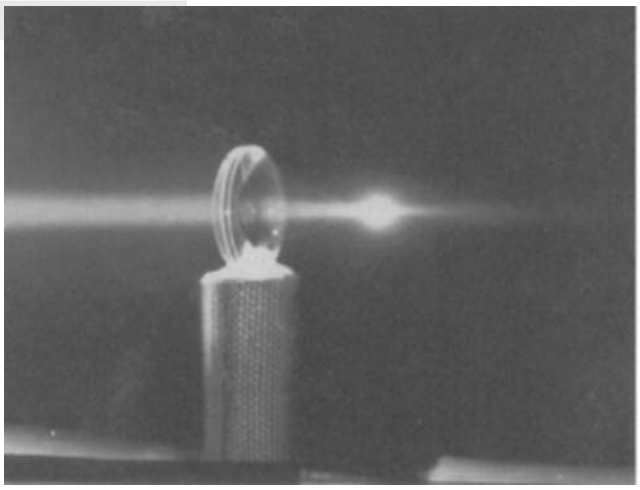


Fig. 18.18 Focusing of a 3 MW peak power pulsed ruby laser beam. At the focus, the electric field strengths are of the order of 10^9 V/m which results in the creation of a spark in the air. (*Photograph courtesy: Dr. R. W. Terhune*).



Fig. 18.19 A focussed laser beam drilling through concrete. Photograph courtesy: Dr. Brahma Nand Upadhyay, RRCAT, Indore. A colour photograph appears as Fig. 40 in prelim pages.

Example 18.7 We consider a 6 cm diameter incandescent source (like a 500 W bulb) at a distance of about 5 m from the eye (see Fig. 18.20). We assume the pupil diameter to be about 2 mm. Thus,

Area of the pupil of the eye $\approx \pi (1 \times 1)$ mm$^2 \approx 3 \times 10^{-6}$ m^2

$$\text{Power entering eye} \approx (500 \text{ W}) \times \frac{\pi r^2}{4\pi R^2} \approx 5 \times 10^{-6} \text{ W}$$

$$\text{Radius of image} = \text{Radius of source} \times \text{demagnification}$$

$$\approx 3 \text{ cm} \times \frac{2.5}{500} \approx 1.5 \times 10^{-4} \text{ m}$$

where we have assumed the image to be formed at a distance of about 2.5 cm from the pupil of the eye. Thus,

The power density in image

$$= \frac{(5\times10^{-6}\text{ W})}{\pi\times(1.5\times10^{-4})^2\text{ m}^2} \approx 70\text{ W/m}^2$$

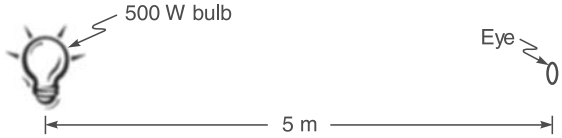


Fig. 18.20 A 500 W bulb at a distance of about 5 m from the eye.

Example 18.8 We next calculate the intensity at the retina if we are directly looking at the Sun (see Fig. 18.21). Now

The intensity of solar energy on earth ≈ 1.35 kW/m^2

Thus, the energy entering the eye

$$\approx 1.35\times10^3\times\pi\times10^{-6}\approx 4\text{ mW}$$

The Sun subtends about 0.5° on the earth. Thus,

The diameter of the image of the Sun

$$\approx 0.5\times\frac{\pi}{180}\times25\approx 0.2\text{ mm}$$

$$= 2\times10^{-4}\text{ m}$$

and

The power density in image

$$\approx \frac{4\times10^{-3}\text{ W}}{\pi\times(2\times10^{-4})^2\text{ m}^2}$$

$$\approx 30\text{ kW/m}^2$$

To summarize, a 2 mW diffraction limited laser beam incident on the eye can produce an intensity of about 10^6 W/m^2 at the retina—this would certainly damage the retina. Thus, whereas it is quite safe to look at a 500 W ordinary light bulb, it is very dangerous to look directly into a 2 mW laser beam. Indeed, because a laser beam can be focused to very narrow areas, it has found important applications in areas like eye surgery, welding, etc.

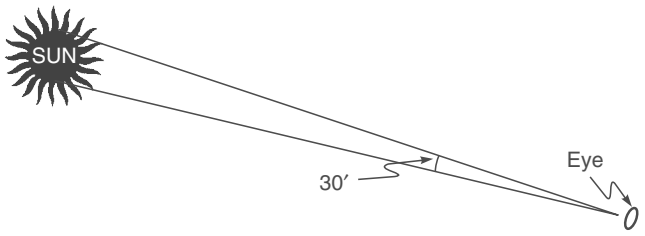


Fig. 18.21 If we look directly at the sun, intensities as high as 130 kW/m^2 are produced; this can damage the retina of the eye!

From the above discussion it immediately follows that greater the radius of the beam, the smaller will be the size of the focused spot and hence greater will be the intensity at the focused spot. Indeed, one may use a beam expander (see Fig. 18.22) to produce a beam of greater size and hence a smaller focused spot size. However, after the focused spot, the beam would have a greater divergence and would therefore expand within a very short distance. One usually defines a *depth of focus* as the distance over which the intensity of the beam (on the axis) decreases by a certain factor of the value at the focal point. Thus, a small focused spot would lead to a small depth of focus. We may mention here that the intensity distribution at the focal plane of the lens is given by Eq. (18.19) where the parameter v is given by Eq. (18.21). On the other hand, the intensity along the axis is given by

$$I = I_0\left[\frac{\sin(w/4)}{w/4}\right]^2 \qquad \textbf{(18.35)}$$

where

$$w = \frac{2\pi}{\lambda}\left(\frac{a}{f}\right)^2 z \qquad \textbf{(18.36)}$$

and $z = 0$ represents the focal plane.*

It can be readily seen that the intensity would drop by about 20% at

$$z \approx \pm 0.5\,\lambda\,(f/a)^2 \qquad \textbf{(18.37)}$$

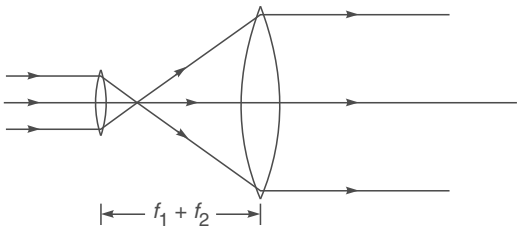


Fig. 18.22 Two convex lenses separated by a distance equal to the sum of their focal lengths act like a beam expander.

which is usually referred to as the depth of the focus or focal tolerance. Notice that larger the value of a, smaller will be the focal tolerance. For $\lambda \approx 6\times10^{-5}$ cm, $f \approx 10$ cm and $a \approx 1$ cm, the focal tolerance is about 3×10^{-3} cm.

18.5 LIMIT OF RESOLUTION LO 5

Consider two point sources, such as stars (so that we can consider plane waves entering the aperture) being focused by a telescope objective of diameter D (see Fig. 18.23). As discussed in the previous section, the system can be thought

* The derivation of the formulae has been given at many places—see, e.g., Sec. 6.5 of Ref. 18.6.

of as being equivalent to a circular aperture of diameter D, followed by a converging lens of focal length f, as shown in Fig. 18.8. As such, each point source will produce its Airy pattern as schematically shown in Fig. 18.23. The diameters of the Airy rings will be determined by the diameter of the objective, its focal length and the wavelength of light (see Example 18.3).

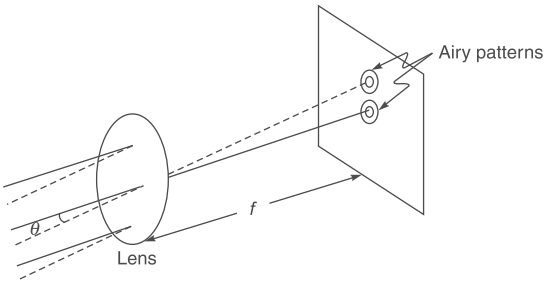


Fig. 18.23 The image of two distant objects on the focal plane of a convex lens. If the diffraction patterns are well separated, they are said to be resolved.

In Fig. 18.23 the Airy patterns are shown to be quite far away from each other and, therefore, the two objects are said to be well resolved. Since the radius of the first ring is $1.22\ \lambda f/D$ the Airy patterns will overlap more for smaller values of D and hence for better resolution one requires a larger diameter of the objective. It is for this reason that a telescope is usually characterized by the diameter of the objective; for example, a 40 inch telescope implies that the diameter of the objective is 40″. In Fig. 18.24 we have shown the image of the binary star Zeta Bootis by a 2.56 m telescope aperture; the Airy disc around each of the stars can be seen.

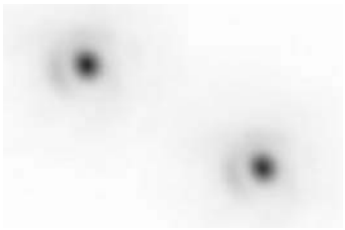


Fig. 18.24 Image of the binary star Zeta Bootis by a 2.56 m telescope aperture; the Airy disc around each of the stars can be seen. The photograph is by Dr. Bob Tubbs; used with permission from Dr. Tubbs. [*A color photograph appears as Fig. 24 in the prelim pages.*]

In Figs. 18.25 and 18.26, we have plotted the independent intensity distributions and their resultant produced by two distant objects for various angular separations; in each case we have assumed that the two sources produce the same intensity at their respective central spots. Obviously, the

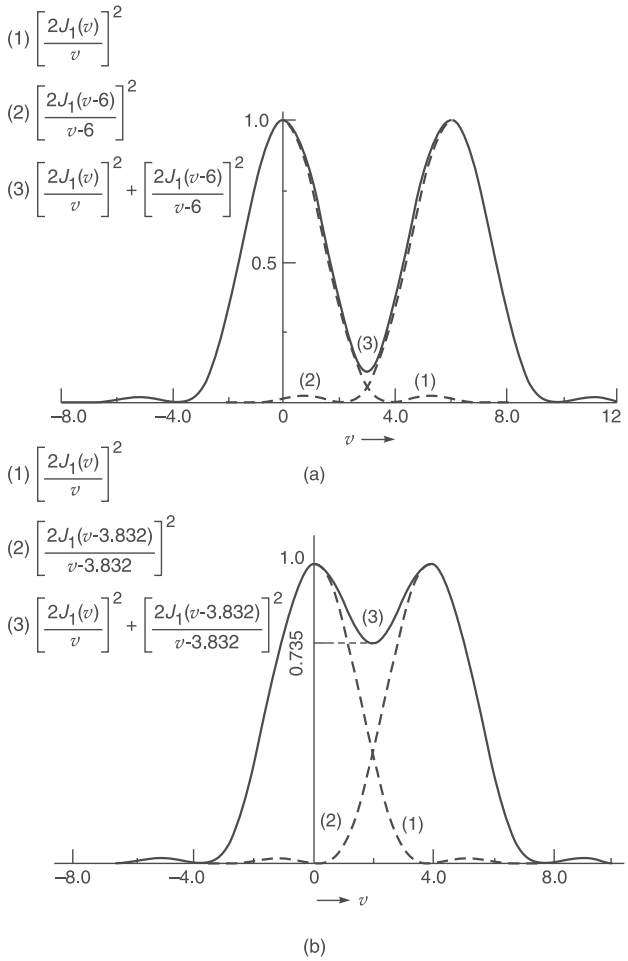


Fig. 18.25 The dashed curves correspond to the intensity distribution produced by two point sources (producing the same intensity at the central spot) independently; the solid curves represent the resultant. (a) and (b) correspond to the angular separation of the two objects equal to $6\lambda/\pi D$ and $1.22\lambda/D$ respectively. In the first case the objects are well resolved and in the second case (according to the Rayleigh criterion) they are just resolved.

resultant intensity distributions are quite complicated (see Fig. 18.27); what we have plotted in Figs. 18.25 and 18.26 are the intensity distributions on the line joining the two centers of the Airy patterns; we should mention here that since the point sources are independent sources, their intensity distributions (Airy patterns) will add. If we choose this line as x-axis then the parameter v in Figs. 18.25 and 18.26 is given by

$$v = \frac{2\pi a}{\lambda f} x \qquad \textbf{(18.38)}$$

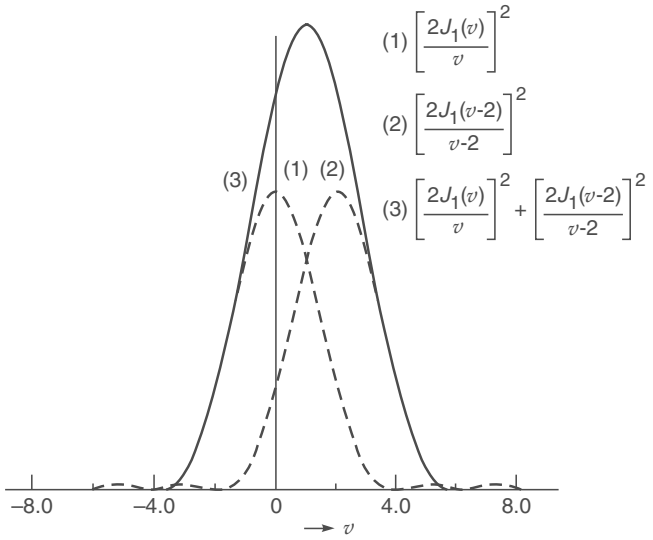


Fig. 18.26 The dashed curves correspond to the intensity distribution produced independently by two distant point objects having an angular separation of $2\lambda/\pi D$. The resultant intensity distribution (shown as a solid curve) has only one peak and hence the objects are unresolved.

Now, the intensity distributions given in Fig. 18.23(a) correspond to two distant point objects having an angular separation of $6\ \lambda/\pi D$ and as can be seen the two images are clearly resolved. Figure 18.26 corresponds to

$$\Delta\theta \simeq \frac{2}{\pi}\frac{\lambda}{D} \qquad \textbf{(18.39)}$$

and as can be seen, the resultant intensity distribution has only one peak and therefore the two points cannot be resolved at all. Finally, if the angular separation of the two objects is $1.22\lambda/D$ then the central spot of the one pattern falls on the first minimum of the second and the objects are said to be just resolved. This criterion of limit of resolution is called the Rayleigh criterion of resolution and the intensity distribution corresponding to this is plotted in Fig. 18.25(b). The actual diffraction patterns are shown in Fig. 18.27.

In order to get a numerical appreciation of the above results we consider a telescope objective whose diameter and focal length are 5 cm and 30 cm, respectively. Assuming the light wavelength to be 6×10^{-5} cm, one finds that the minimum angular separation of two distant objects which can just be resolved will be

$$\frac{1.22\lambda}{D} = \frac{1.22 \times 6 \times 10^{-5}}{5} \simeq 1.5 \times 10^{-5} \text{ radians}$$

Further, the radius of the first dark ring (of the Airy pattern) will be

$$\frac{1.22\lambda}{D} \times \text{focal length} = \frac{1.22 \times 6 \times 10^{-5}}{5} \times 30$$

$$\simeq 4.5 \times 10^{-4} \text{ cm}$$

It is immediately obvious that the larger the diameter of the objective, the better will be its resolving power. For example, the diameter of the largest telescope objective is about 80″ and the corresponding angular separation of the objects that it can resolve is $\simeq$ 0.07 sec of arc. This very low limit of resolution is never achieved in ground based telescopes due to the turbulence of the atmosphere. However, a larger aperture still provides a larger light gathering power and hence the ability to see deeper in space.

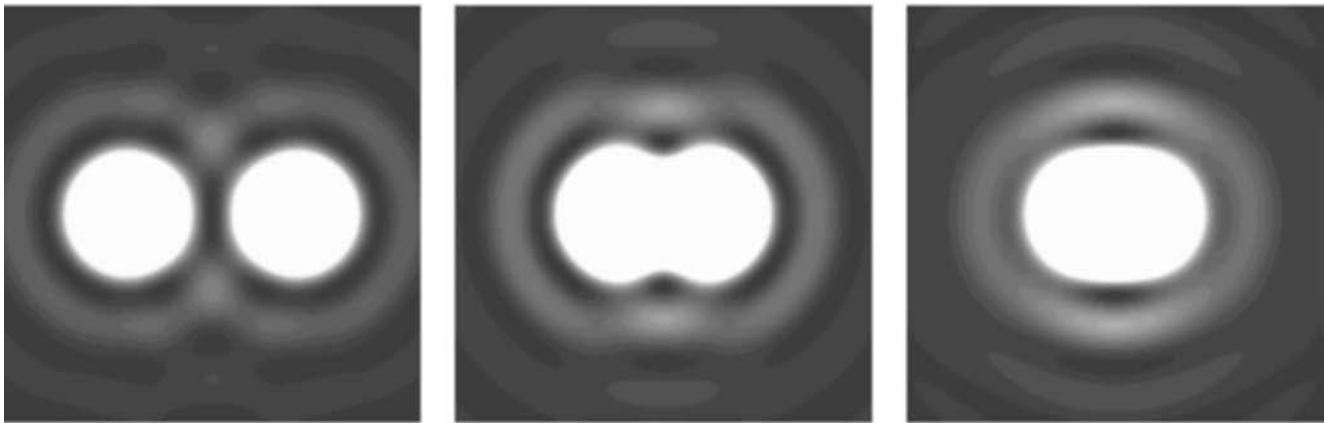


Fig. 18.27 Computer generated intensity distributions corresponding to two point sources when they are: (a) well resolved, (b) just resolved, and (c) unresolved.

It is of interest to note that if we assume that the angular resolution of the human eye is primarily due to diffraction effects then it will be given by

$$\Delta\theta \sim \frac{\lambda}{D} \approx \frac{6\times10^{-5}}{2\times10^{-1}} = 3\times10^{-4}\ \text{rad.} \qquad \textbf{(18.40)}$$

where we have assumed the pupil diameter to be 2 mm. Thus, at a distance of 20 m, the eye should be able to resolve two points which are separated by a distance

$$3\times10^{-4}\times20 = 6\times10^{-3}\ \text{m} = 6\ \text{mm}$$

One can indeed verify that this result is qualitatively valid by finding the distance at which the millimetre scale will become blurred.

In the above discussion, we have assumed that the two object points produce identical (but displaced) Airy patterns. If that is not the case then the two central maxima will have different intensities; accordingly one has to set up a modified criterion for the limit of resolution such that the two maxima stand out.

18.5.1 Resolving Power of a Microscope

We next consider the resolving power of a microscope objective of diameter D as shown in Fig. 18.28. Let P and Q represent two closely spaced self-luminous point objects which are to be viewed through the microscope. Assuming the absence of any geometrical aberrations, rays emanating from the points P and Q will produce spherical wavefronts (after refraction through the lens) which will form Airy patterns around their paraxial image points P' and Q'. For the points P and Q to be just resolved, the point Q' should lie on the first dark ring surrounding the point P' and, therefore, we must have

$$\sin\alpha' \approx \frac{1.22\,\lambda}{D} = \frac{1.22\,\lambda_0}{n'D} \qquad \textbf{(18.41)}$$

where n and n' represent the refractive indices of the object and image spaces, λ_0 and $\lambda\,(=\lambda_0/n')$ represent the wavelength of light in free space and in the medium of refractive index n' respectively. The angle α' is defined in Fig. 18.28 and we have

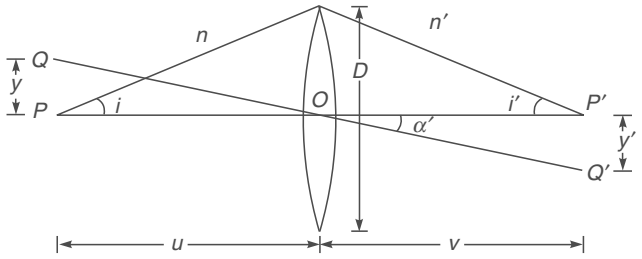


Fig. 18.28 The resolving power of a microscope objective.

$$\sin\alpha' \approx \frac{y'}{OP'} = \frac{y'\tan i'}{D/2} \approx \frac{y'\sin i'}{D/2} \qquad \textbf{(18.42)}$$

where we have assumed $\sin i' \approx \tan i'$, this is justified since the image distance (OP') is large compared to D. Using Eqs. (18.41) and (18.42), we get

$$y' \approx \frac{0.61\,\lambda_0}{n'\sin i'}$$

If we now use the sine law $n'y'\sin i' = ny\sin i$ [see Eq. (4.39) of Chapter 4], we get

$$y \approx \frac{0.61\,\lambda_0}{n\sin i} \qquad \textbf{(18.43)}$$

which represents the smallest distance that the microscope can resolve. The quantity $n\sin i$ is the numerical aperture of the optical system and the resolving power increases with increase in the numerical aperture. It is for this reason that in some microscopes the space between the object and the objective is filled with an oil—and they are referred to as 'oil immersion objectives'. Equation (18.43) also tells us that the resolving power increases with decrease in λ. As such, one often uses blue light (or even ultraviolet light) for the illumination of the object. For example, in an electron microscope the de Broglie wavelength of electrons accelerated to 100 keV is about 0.03×10^{-8} cm and therefore such a microscope has a very high resolving power.

In the above analysis, we have assumed that the two object points are self-luminous so that the intensities can be added up. However, in actual practice, the objects are illuminated by the same source and, therefore, in general, there is some phase relationship between the waves emanating from the two object points; for such a case the intensities will not be strictly additive (see Sec. 14.6), nevertheless Eq. (18.43) will give the correct order for the limit of resolution.

18.6 TWO-SLIT FRAUNHOFER DIFFRACTION PATTERN

LO 6

In Sec. 18.3, we had studied the Fraunhofer diffraction pattern produced by a slit of width b and had found that the intensity distribution consisted of maxima and minima. In this section, we will study the Fraunhofer diffraction pattern produced by two parallel slits (each of width b) separated by a distance d. We would find that the resultant intensity distribution is a product of the single-slit diffraction pattern and the interference pattern produced by two point sources separated by a distance d.

In order to calculate the diffraction pattern we use a method similar to that used for the case of a single slit and assume that the slits consist of a large number of equally spaced point sources and that each point on the slit is a source of Huygens' secondary wavelets. Let the point sources be at $A_1, A_2, A_3, \ldots$ (in the first slit) and at $B_1, B_2, B_3, \ldots$ (in the second slit) [see Fig. 18.29]. As before, we assume that the distance between two consecutive points in either of the slits is Δ. If the diffracted rays make an angle θ with the normal to the plane of the slits, then the path difference between the disturbances reaching the point P from two consecutive points in a slit will be $\Delta \sin \theta$. The field produced by the first slit at the point P will, therefore, be given by [see Eq. (18.9)]

$$E_1 = A \frac{\sin \beta}{\beta} \cos (\omega t - \beta)$$

Similarly, the second slit will produce a field

$$E_2 = A \frac{\sin \beta}{\beta} \cos (\omega t - \beta - \Phi_1)$$

at the point P, where

$$\Phi_1 = \frac{2\pi}{\lambda} d \sin \theta$$

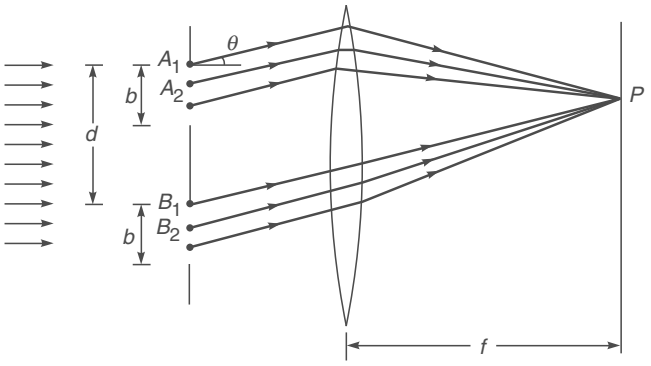


Fig. 18.29 Fraunhofer diffraction of a plane wave incident normally on a double slit.

represents the phase difference between the disturbances (reaching the point P) from two corresponding points on the slits; by corresponding points we imply pairs of points like (A_1, B_1), $(A_2, B_2), \ldots$ which are separated by a distance d. Hence the resultant field will be

$$E = E_1 + E_2$$

$$= A \frac{\sin \beta}{\beta} [\cos (\omega t - \beta) + \cos (\omega t - \beta - \Phi_1)]$$

which represents the interference of two waves, each of amplitude $A \frac{\sin \beta}{\beta}$ and differing in phase by Φ_1. The above equation can be rewritten in the form

$$E = A \frac{\sin \beta}{\beta} \cos \gamma \cos \left(\omega t - \frac{1}{2} \beta - \frac{1}{2} \Phi_1 \right)$$

where

$$\gamma = \frac{\Phi_1}{2} = \frac{\pi}{\lambda} d \sin \theta \qquad \textbf{(18.44)}$$

The intensity distribution will be of the form

$$I = 4I_0 \frac{\sin^2 \beta}{\beta^2} \cos^2 \gamma \qquad \textbf{(18.45)}$$

where $I_0 \sin^2 \beta/\beta^2$ represents the intensity distribution produced by one of the slits. As can be seen, the intensity distribution is a product of two terms; the first term ($\sin^2 \beta/\beta^2$) represents the diffraction pattern produced by a single slit of width b and the second term ($\cos^2 \gamma$) represents the interference pattern produced by two point sources separated by a distance d. Indeed, if the slit widths are very small (so that there is almost no variation of the $\sin^2 \beta/\beta^2$ term with θ) then one simply obtains the Young's interference pattern (see Sec. 14.6).

In Fig. 18.30, we have shown the two slit diffraction patterns corresponding to d = 0, 0.0176, 0.035 and 0.070 cm with b = 0.0088 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The intensity distribution as predicted by Eq. (18.45) is shown in Fig. 18.31.

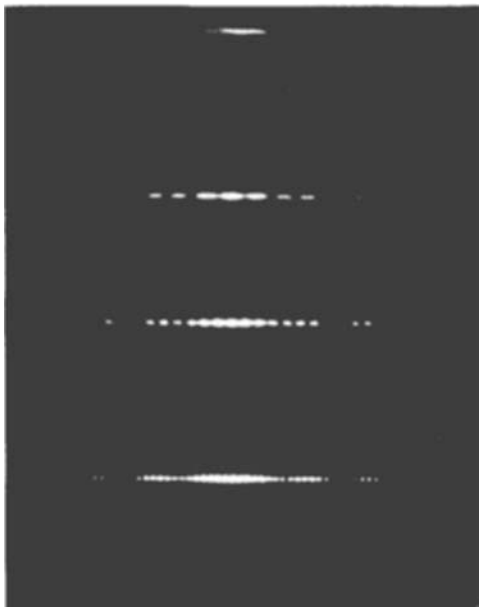


Fig. 18.30 The double-slit Fraunhofer diffraction pattern corresponding to b = 0.0088 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The values of d are 0, 0.0176, 0.035 and 0.070 cm respectively [Adapted from Ref. 17; used with permission].

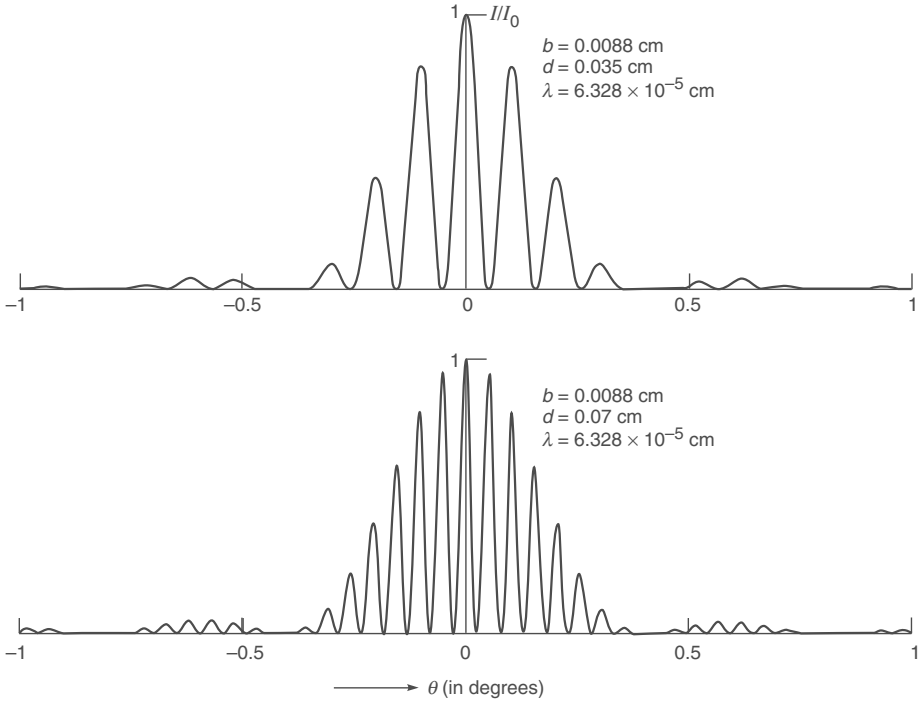


Fig. 18.31 The double-slit intensity distribution as predicted by Eq. (18.45) corresponding to $d = 0.035$ cm and 0.070 cm, respectively (b = 0.0088 cm and $\lambda = 6.328 \times 10^{-5}$ cm).

18.6.1 Positions of Maxima and Minima

Equation (45) tells us that the intensity is zero wherever

$$\beta = \pi, 2\pi, 3\pi, \ldots$$

or when

$$\gamma = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$$

The corresponding angles of diffraction will be given by the following equation:

$$\left.\begin{aligned} b \sin\theta &= m\lambda; \quad (m = 1, 2, 3, \ldots) \\ \text{and} \qquad d \sin\theta &= \left(n + \frac{1}{2}\right)\lambda; \quad (n = 1, 2, 3, \ldots) \end{aligned}\right\} \qquad \textbf{(18.46)}$$

The interference maxima occur when

$$\gamma = 0, \pi, 2\pi, \ldots$$

or when,

$$d \sin\theta = 0, \lambda, 2\lambda, 3\lambda, \ldots \qquad \textbf{(18.47)}$$

The actual positions of the maxima will approximately occur at the above angles provided the variation of the diffraction term is not too rapid. Further, a maximum may not occur at all if θ corresponds to a diffraction minimum, i.e., if $b \sin\theta = \lambda, 2\lambda, 3\lambda, \ldots$ These are usually referred to as missing orders. For example, in Fig. 18.31 we can see that for b = 0.0088 cm, the interference maxima are extremely weak around $\theta \simeq 0.41^\circ$; this is because of the fact that at

$$\theta = \sin^{-1}\left(\frac{\lambda}{b}\right)$$

$$= \sin^{-1}\left[\frac{6.328 \times 10^{-5}}{8.8 \times 10^{-3}}\right] = \sin^{-1}\left[7.19 \times 10^{-3}\right]$$

$$\simeq 0.00719 \text{ radians}$$

$$\simeq 0.412^\circ$$

the first minimum of the diffraction term occurs.

Example 18.9 Consider the case when $b = 8.8 \times 10^{-3}$ cm, $d = 7.0 \times 10^{-2}$ cm and $\lambda = 6.328 \times 10^{-5}$ cm (see Fig. 18.31). How many interference minima will occur between the two diffraction minima on either side of the central maximum? In the experimental arrangement corresponding to Fig. 18.30 the screen was placed at a distance of 15′. Calculate the fringe width.

Solution: The interference minima will occur when Eq. (18.46) is satisfied, i.e., when

$$\sin\theta = \left(n + \frac{1}{2}\right)\frac{\lambda}{d} = 0.904 \times 10^{-3}\left(n + \frac{1}{2}\right);$$

$$n = 0, 1, 2, \ldots$$

$$= 0.452 \times 10^{-3}, 1.356 \times 10^{-3}, 2.260 \times 10^{-3},$$

$$3.164 \times 10^{-3}, 4.068 \times 10^{-3}, 4.972 \times 10^{-3},$$

$$5.876 \times 10^{-3}, 6.780 \times 10^{-3}$$

Thus, there will be sixteen minima between the two first order diffraction minima.

The angular separation between two interference maxima is approximately given by (see Eq. 18.47)

$$\Delta\theta \simeq \frac{\lambda}{d} = 0.904 \times 10^{-4}$$

Thus, the fringe width will be

$$15 \times 12 \times 2.54 \times 0.904 \times 10^{-4} \simeq 0.0413 \text{ cm}$$

18.7 N-SLIT FRAUNHOFER DIFFRACTION PATTERN

LO 6

We next consider the diffraction pattern produced by N parallel slits, each of width b; the distance between two consecutive slits is assumed to be d.

As before, we assume that each slit consists of n equally spaced point sources with spacing Δ (see Fig. 18.32). Thus, the field at an arbitrary point P will essentially be a sum of N terms:

$$E = A\frac{\sin\beta}{\beta}\cos(\omega t - \beta) + A\frac{\sin\beta}{\beta}\cos(\omega t - \beta - \Phi_1) + \ldots + A\frac{\sin\beta}{\beta}\cos(\omega t - \beta - (N-1)\Phi_1) \quad \textbf{(18.48)}$$

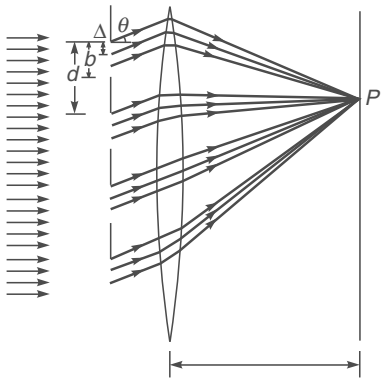


Fig. 18.32 Fraunhofer diffraction of a plane wave incident normally on a multiple slit.

where the first term represents the amplitude produced by the first slit, the second term by the second slit, etc., and the various symbols have the same meaning as in Sec. 18.5. Rewriting Eq. (18.48) we get

$$E = \frac{A\sin\beta}{\beta}\left[\cos(\omega t - \beta) + \cos(\omega t - \beta + \Phi_1) + \ldots + \cos(\omega t - \beta - (N-1)\Phi_1)\right]$$

$$= \frac{A\sin\beta}{\beta}\frac{\sin N\gamma}{\sin\gamma}\cos\left[\omega t - \beta - \frac{1}{2}(N-1)\Phi_1\right] \quad \textbf{(18.49)}$$

where

$$\gamma = \frac{\Phi_1}{2} = \frac{\pi}{\lambda} d\sin\theta$$

The corresponding intensity distribution will be

$$I = I_0\frac{\sin^2\beta}{\beta^2}\frac{\sin^2 N\gamma}{\sin^2\gamma} \quad \textbf{(18.50)}$$

where $I_0 \sin^2\beta/\beta^2$ represents the intensity distribution produced by a single slit. As can be seen, the intensity distribution is a product of two terms; the first term $\left(\frac{\sin^2\beta}{\beta^2}\right)$ represents the diffraction pattern produced by a single slit and the second term $\left(\frac{\sin^2 N\gamma}{\sin^2\gamma}\right)$ represents the interference pattern produced by N equally spaced point sources. For $N = 1$, Eq. (18.50) reduces to the single-slit diffraction pattern [see Eq. (18.10)] and for $N = 2$, to the double slit diffraction pattern [see Eq. (18.45)]. In Figs. 18.33 and 18.34, we have given a plot of the function

$$\frac{\sin^2 N\gamma}{\sin^2\gamma}$$

as a function of γ for $N = 5$ and $N = 12$. One can immediately see that as the value of N becomes very large, the above function would become very sharply peaked at $\gamma = 0, \pi, 2\pi, \ldots$ Between the two peaks, the function vanishes when

$$\gamma = \frac{p\pi}{N}; \quad p = \pm 1, \pm 2, \ldots \quad \text{but} \quad p \neq 0, \pm N, \pm 2N$$

which are referred to as secondary minima.

18.7.1 Positions of Maxima and Minima

When the value of N is very large, one obtains intense maxima at $\gamma \simeq m\pi$, i.e., when

$$d\sin\theta = m\lambda \quad (m = 0, 1, 2, \ldots) \quad \textbf{(18.51)}$$

This can be easily seen by noting that

$$\underset{\gamma\to m\pi}{Lt}\frac{\sin N\gamma}{\sin\gamma} = \underset{\gamma\to m\pi}{Lt}\frac{N\cos N\gamma}{\cos\gamma} = \pm N;$$

thus, the resultant amplitude and the corresponding intensity distributions are given by

$$E = N\frac{A\sin\beta}{\beta} \quad \textbf{(18.52)}$$

and

$$I = N^2 I_0\frac{\sin^2\beta}{\beta^2} \quad \textbf{(18.53)}$$

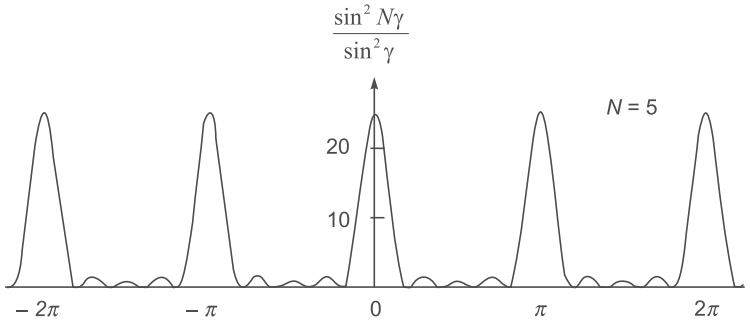


Fig. 18.33 The variation of the function $\sin^2(N\gamma)/\sin^2\gamma$ with γ for $N = 5$.

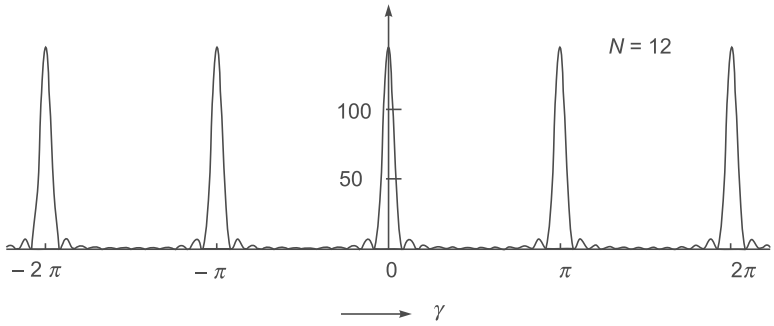


Fig. 18.34 The variation of the function $\sin^2(N\gamma)/\sin^2\gamma$ with γ for $N = 12$. As N becomes larger, the function would become more and more sharply peaked at $\gamma = 0, \pm\pi, \pm 2\pi, \pm 3\pi, \ldots$.

where

$$\beta = \frac{\pi b \sin\theta}{\lambda} = \frac{\pi b}{\lambda}\frac{m\lambda}{d} = \frac{\pi b m}{d} \qquad \textbf{(18.54)}$$

Such maxima are known as **principal maxima**. Physically, at these maxima the fields produced by each of the slits are in phase and, therefore, they add up and the resultant field is N times the field produced by each of the slits; consequently, the intensity has a large value unless $\frac{\sin^2\beta}{\beta^2}$ itself is very small. Since $|\sin\theta| \leq 1$, m cannot be greater than d/λ [see Eq. (18.51)]; thus, there will only be a finite number of principal maxima.

From Eq. (18.50) it can easily be seen that the intensity is zero when either

$$b \sin\theta = n\lambda,\ n = 1, 2, 3, \ldots \qquad \textbf{(18.55)}$$

or

$$N\gamma = p\pi,\ p \neq N, 2N, \ldots \qquad \textbf{(18.56)}$$

Equation (18.55) gives us the minima corresponding to the single slit diffraction pattern. The angles of diffraction corresponding to Eq. (18.56) are

$$d \sin\theta = \frac{\lambda}{N}, \frac{2\lambda}{N}, \ldots, \frac{(N-1)\lambda}{N}, \frac{(N+1)\lambda}{N}, \frac{(N+2)\lambda}{N},$$

$$\ldots, \frac{(2N-1)\lambda}{N}, \frac{(2N+1)\lambda}{N}, \frac{(2N+2)\lambda}{N}, \ldots \qquad \textbf{(18.57)}$$

Thus, between two principal maxima we have $(N-1)$ minima. Between two such consecutive minima the intensity has to have a maximum; these maxima are known as secondary maxima. Typical diffraction patterns for $N = 1$, 2, 3, and 4 are shown in Fig. 18.35 and the intensity distribution as predicted by Eq. (18.50) for $N = 4$ is shown in Fig. 18.36. When N is very large the principal maxima will be much more intense in comparison to the secondary maxima. We may mention here two points:

(a) A particular principal maximum may be absent if it corresponds to the angle which also determines the minimum of the single-slit diffraction pattern. This will happen when

$$d \sin\theta = m\lambda \qquad \textbf{(18.58)}$$

and

$$b \sin\theta = \lambda, 2\lambda, 3\lambda, \ldots \qquad \textbf{(18.59)}$$

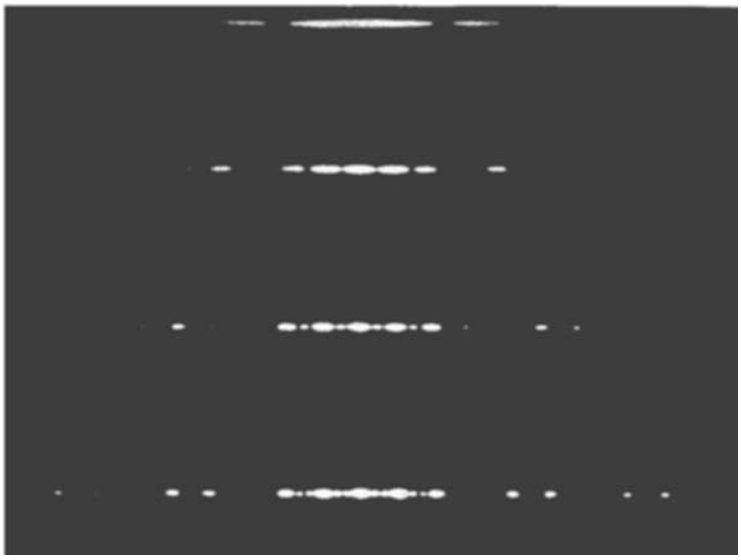


Fig. 18.35 The multiple-slit Fraunhofer diffraction patterns corresponding to b = 0.0044 cm, d = 0.0132 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The number of slits are 1, 2, 3 and 4 respectively (Adapted from Ref. 18.17; used with permission).

are satisfied simultaneously and is usually referred to as a missing order. Even when Eq. (18.59) does not hold exactly (i.e., if $b \sin\theta$ is close to an integral multiple of λ), the intensity of the corresponding principal maximum will be very weak (see, for example, Fig. 18.36 around $\theta \approx 0.8°$).

(b) In addition to the minima predicted by Eq. (18.56), we will also have the diffraction minima (see Eq. 18.55); however, when N is very large, the number of such minima will be very small.

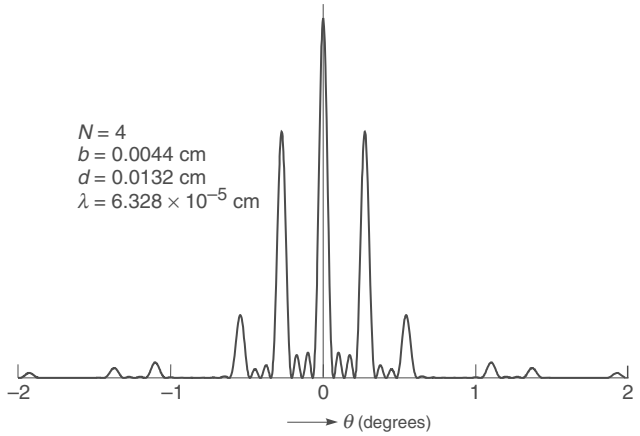


Fig. 18.36 The intensity distribution corresponding to the four-slit Fraunhofer diffraction pattern as predicted by Eq. (18.50) corresponding to b = 0.0044 cm, d = 0.0132 cm and $\lambda = 6.328 \times 10^{-5}$ cm. The principle maxima occur at $\theta \approx 0.275°$, $0.55°$, $0.82°$, $1.1°$, ... Notice the (almost) absent third order.

18.7.2 Width of the Principal Maxima

We have shown above that in the diffraction pattern produced by N slits, the mth order principal maximum occurs at

$$d \sin\theta_m = m\lambda, m = 0, 1, 2, \ldots \quad \textbf{(18.60)}$$

Further, the minima occur at the angles given by Eq. (18.57). If $\theta_m + \Delta\theta_{1m}$ and $\theta_m - \Delta\theta_{2m}$ represent the angles of diffraction corresponding to the first minimum on either side of the principal maximum, then $\frac{1}{2}(\Delta\theta_{1m} + \Delta\theta_{2m})$ is known as the **angular half width of the *m*th order principal maximum**. For a large value of N, $\Delta\theta_{1m} \simeq \Delta\theta_{2m}$ which we write as $\Delta\theta_m$. Clearly,

$$d \sin(\theta_m \pm \Delta\theta_m) = m\lambda \pm \frac{\lambda}{N} \quad \textbf{(18.61)}$$

But

$$\sin(\theta_m \pm \Delta\theta_m) = \sin\theta_m \cos\Delta\theta_m \pm \cos\theta_m \sin\Delta\theta_m$$

$$\simeq \sin\theta_m \pm \Delta\theta_m \cos\theta_m \quad \textbf{(18.62)}$$

Thus Eq. (18.61) gives us

$$\Delta\theta_m \simeq \frac{\lambda}{Nd\cos\theta_m} \quad \textbf{(18.63)}$$

which shows that the principal maximum becomes sharper as N increases.

18.8 THE DIFFRACTION GRATING

LO 7

In Sec. 18.6, we have discussed the diffraction pattern produced by a system of parallel equidistant slits. An arrangement which essentially consists of a large number of equidistant slits is known as a **diffraction grating**; the corresponding diffraction pattern is known as the grating spectrum. Since the exact positions of the principal maxima in the diffraction pattern depend on the wavelength, the principal maxima corresponding to different spectral lines (associated with a source) will correspond to different angles of diffraction. Thus the grating spectrum provides us with an easily obtainable experimental set up for determination of wavelengths. From Eq. (18.63) we see that for narrow principal maxima (i.e., sharper spectral lines), a large value of N is required. A good quality grating, therefore, requires a large number of slits (typically about 15,000 per inch). This is achieved by ruling grooves with a diamond point on an optically transparent sheet of material; the grooves act as opaque spaces. After each groove is ruled, the machine lifts the diamond point and moves the sheet forward for the ruling of the next groove. Since the distance between two consecutive grooves is extremely small, the movement of the sheet is obtained with the help of the rotation of a screw which drives the carriage carrying it. Further, one of the important requirements of a good quality grating is that the lines should be as equally spaced as possible; consequently, the pitch of the screw must be constant, and it was not until the manufacture of a nearly perfect screw (which was achieved by Rowland in 1882) that the problem of construction of gratings was successfully solved. Rowland's arrangement gave 14,438 lines per inch, corresponding to $d = 2.54/14438 = 1.759 \times 10^{-4}$ cm. For such a grating, for $\lambda = 6 \times 10^{-5}$ cm, the maximum value of m would be 2, and, therefore, only the first two orders of the spectrum will be observed. However, for $\lambda = 5 \times 10^{-5}$ cm, the third order spectrum will also be visible.

Commercial gratings are produced by taking the cast of an actual grating on a transparent film like that of cellulose acetate. An appropriate strength solution of cellulose acetate is poured on the ruled surface and allowed to dry to form a strong thin film, detachable from the parent grating. These impressions of a grating are preserved by mounting the film between two glass sheets. Nowadays, gratings are also produced holographically, where one records the interference pattern between two plane or spherical waves (see Example 14.5). In contrast to ruled gratings, holographic gratings have a much larger number of lines/cm.

18.8.1 The Grating Spectrum

In Sec. 18.6, we have shown that the positions of the principal maxima are given by

$$d \sin \theta = m\lambda; \quad m = 0, 1, 2, \ldots \qquad \textbf{(18.64)}$$

This relation, which is also called the **grating equation**, can be used to study the dependence of the angle of diffraction θ on the wavelength λ. The zeroeth order principal maximum occurs at $\theta = 0$ irrespective of the wavelength. Thus, if we are using a polychromatic source (e.g., white light) then the central maximum will be of the same colour as the source itself. However, for $m \neq 0$, the angles of diffraction are different for different wavelengths and, therefore, various spectral components appear at different positions. Thus, by measuring the angles of diffraction for various colors one can (knowing the value of m) determine the values of the wavelengths. It may be mentioned that the intensity is maximum for the zeroeth order spectrum (where no dispersion occurs) and it falls off as the value of m increases.

If we differentiate Eq. (18.64), we would obtain

$$\frac{\Delta\theta}{\Delta\lambda} = \frac{m}{d \cos\theta} \qquad \textbf{(18.65)}$$

From this result we can deduce the following conclusions:

(a) Assuming θ to be very small (i.e., $\cos\theta \simeq 1$) we can see that the angle $\Delta\theta$ is directly proportional to the order of spectrum (m) for a given $\Delta\lambda$, so that for a given m, $\Delta\theta/\Delta\lambda$ is a constant. Such a spectrum is known as a **normal spectrum** and in this the difference in angle for two spectral lines is directly proportional to the difference in wavelengths. However, for large θ, it can easily be shown that the dispersion is greater at the red end of the spectrum.

(b) Equation (18.65) tells us that $\Delta\theta$ is inversely proportional to d, and therefore smaller the grating element, the larger will be the angular dispersion.

Figures 18.37 and 18.38 show schematic diagrams of the experimental arrangement for studying the grating spectrum of a polychromatic source. In Fig. 18.37, we have shown a small hole placed at the focal plane of the lens L_1. A parallel beam of white light emerging from L_1 falls on the grating and the diffraction pattern is observed on the focal plane of the lens L_2. If instead of a hole we have a slit at the focal plane of L_1 (see Fig. 18.38) — as it is indeed the case in a typical laboratory set up — we would have parallel beams propagating in different directions, and in the focal plane of the lens L_2 we will have a band spectrum as shown in Fig. 18.38.

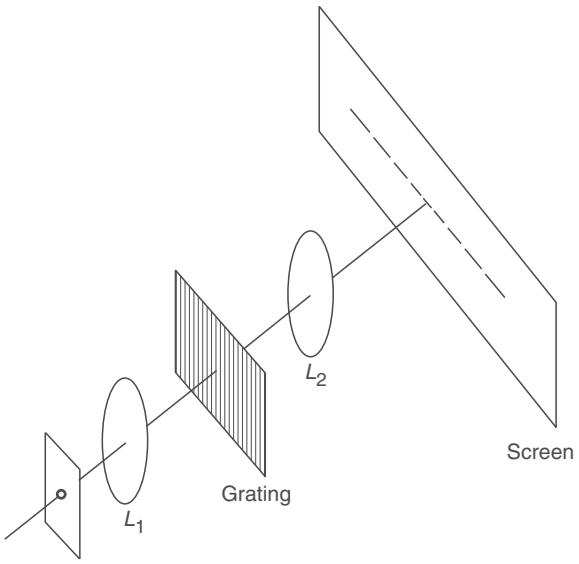


Fig. 18.37 Fraunhofer diffraction of a plane wave incident normally on a grating.

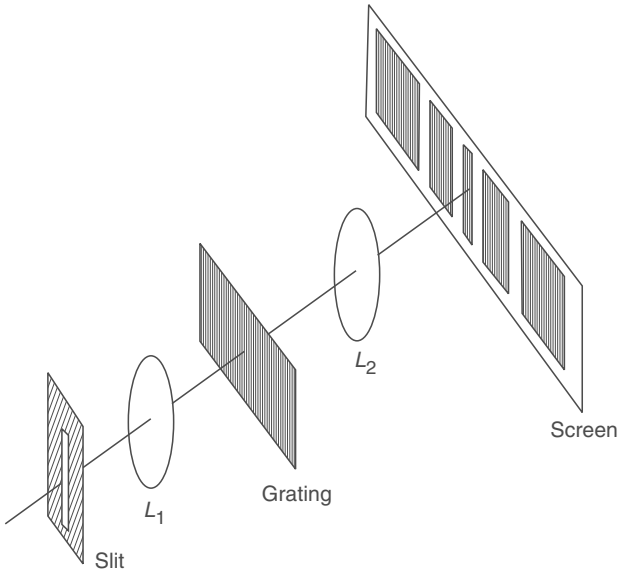


Fig. 18.38 If instead of a point source we have a slit in the focal plane of L_1 then one will obtain bands on the focal plane of L_2.

The lens L_2 is the objective of a telescope and the diffraction pattern is viewed through an eyepiece. The angles of diffraction for various orders of the grating spectrum can be measured and knowing the value of d, one can calculate the wavelength of different spectral lines.

Example 18.10 Consider a diffraction grating with 15,000 lines per inch. **(a)** Show that if we use a white light source the second and third order spectra overlap. **(b)** What will be the angular separation of the D_1 and D_2 lines of sodium in the second order spectra?

Solution: **(a)** The grating element is

$$d = \frac{2.54}{15000} = 1.69 \times 10^{-4} \text{ cm}$$

Let θ_{mv} and θ_{mr} represent the angles of diffraction for the mth order spectrum corresponding to the violet and red colors, respectively. Thus,

$$\theta_{2v} = \sin^{-1} \frac{2 \times 4 \times 10^{-5}}{1.69 \times 10^{-4}} \simeq \sin^{-1} 0.473 \simeq 28.2^\circ$$

$$\theta_{2r} = \sin^{-1} \frac{2 \times 7 \times 10^{-5}}{1.69 \times 10^{-4}} \simeq \sin^{-1} 0.828 \simeq 55.90^\circ$$

and

$$\theta_{3v} = \sin^{-1} \frac{3 \times 4 \times 10^{-5}}{1.69 \times 10^{-4}} \simeq \sin^{-1} 0.710 \simeq 45.23^\circ$$

where we have assumed the wavelengths of the violet and red colours to be 4×10^{-5} cm, and 7×10^{-5} cm, respectively. Since $\theta_{2r} > \theta_{3v}$, the second and third order spectra will overlap. Further since sin $\theta_{3r} > 1$, the third order spectrum for the red colour will not be observed.

(b) Since $d \sin \theta = m\lambda$, we have for small $\Delta\lambda$:

$$(d \cos \theta)\, \Delta\theta = m(\Delta\lambda)$$

or

$$\Delta\theta = \frac{m\Delta\lambda}{d\left\{1 - \left(\frac{m\lambda}{d}\right)^2\right\}^{1/2}}$$

$$\simeq \frac{2 \times 6 \times 10^{-8}}{1.69 \times 10^{-4}\left[1 - \left(\frac{2 \times 6 \times 10^{-5}}{1.69 \times 10^{-4}}\right)^2\right]^{1/2}}$$

$$\simeq 0.0010 \text{ radians} \simeq 3.47'$$

Thus, if we are using telescope of angular magnification 10, the two lines will appear to have an angular separation of 34.7′.

18.8.2 Resolving Power of a Grating

In the case of a grating the resolving power refers to the power of distinguishing two nearby spectral lines and is defined by the following equation:

$$R = \frac{\lambda}{\Delta\lambda} \qquad \textbf{(18.66)}$$

where $\Delta\lambda$ is the separation of two wavelengths which the grating can just resolve; the smaller the value of $\Delta\lambda$, the larger the resolving power.

The Rayleigh criterion (see Sec. 18.4) can again be used to define the limit of resolution. According to this criterion, if the principal maximum corresponding to the wavelength $\lambda + \Delta\lambda$ falls on the first minimum (on the either side of the principal maximum) of the wavelength λ, then the two wavelengths λ and $\lambda + \Delta\lambda$ are said to be just resolved (see Fig. 18.39). If this common diffraction angle is represented by θ and if we are looking at the mth order spectrum, then the two wavelengths λ and $\lambda + \Delta\lambda$ will be just resolved if the following two equations are simultaneously satisfied:

$$d \sin \theta = m(\lambda + \Delta\lambda) \tag{18.67}$$

and

$$d \sin \theta = m\lambda + \frac{\lambda}{N} \tag{18.68}$$

Thus,

$$R = \frac{\lambda}{\Delta\lambda} = mN \tag{18.69}$$

which implies that the resolving power depends on the total number of lines in the grating—obviously on only those lines which are exposed to the incident beam (see the derivation in Sec. 18.6). Further, the resolving power is proportional to the order of the spectrum. Thus, to resolve the D_1 and D_2 lines of sodium ($\Delta\lambda$ = 6 Å) in the first order, N must be at least $(5.89 \times 10^{-5})/(6 \times 10^{-8}) \approx 1{,}000$.

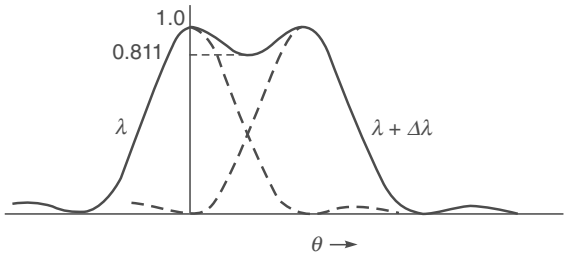


Fig. 18.39 The Rayleigh criterion for the resolution of two spectral lines.

From Eq. (18.69) it appears that the resolving power of the grating would increase indefinitely if N is increased; however, for a given width of the grating $D(= Nd)$, as N is increase, d decreases and therefore the maximum value of m also decreases. Thus if d becomes 2.5λ, only first and second order spectra will be seen and if it is further reduced to about 1.5λ then only the first order spectrum will be seen.

18.8.3 Resolving Power of a Prism

We conclude this section by calculating the resolving power of a prism. Figure 18.40 gives a schematic description of the experimental arrangement for observing the prism spectrum which is determined through the following formula:

$$n(\lambda) = \frac{\sin \dfrac{A + \delta(\lambda)}{2}}{\sin \dfrac{A}{2}} \tag{18.70}$$

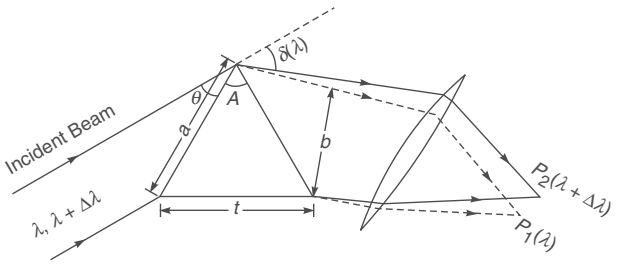


Fig. 18.40 The schematic of the experimental arrangement to observe the prism spectrum. P_1 and P_2 represent the images corresponding to λ and $\lambda + \Delta\lambda$, respectively.

where A represents the angle of the prism and δ the angle of minimum deviation. We assume that the refractive index decreases with λ (which is usually the case) so that δ also decreases with λ. In Fig. 18.40, the points P_1 and P_2 represent the images corresponding to λ and $\lambda + \Delta\lambda$, respectively. We are assuming that $\Delta\lambda$ is small so that the same position of the prism corresponds to the minimum deviation position for both wavelengths. In an actual experiment one usually has a slit source (perpendicular to the plane of the paper) forming line images at P_1 and P_2. Since the faces of the prism are rectangular, the intensity distribution will be similar to that produced by a slit of width b (see Sec. 18.2)*. For the lines to be just resolved the first diffraction minimum [m = 1 in Eq. (18.12)] of λ should fall at the central maximum of $\lambda + \Delta\lambda$, thus we must have

$$\Delta\delta \approx \frac{\lambda}{b} \tag{18.71}$$

In order to express $\Delta\delta$ in terms of $\Delta\lambda$, we differentiate Eq. (18.70):

$$\frac{dn}{d\lambda} = \frac{1}{\sin \dfrac{A}{2}} \cos\left[\frac{A + \delta(\lambda)}{2}\right] \frac{1}{2} \frac{d\delta}{d\lambda}$$

Thus,

$$\Delta\delta = \frac{2 \sin \dfrac{A}{2}}{\cos \dfrac{A + \delta(\lambda)}{2}} \frac{dn}{d\lambda} \Delta\lambda$$

* Since we have a slit source we need not consider the diffraction in a direction perpendicular to the plane of the diagram.

Now from Fig. 18.40, we have

$$\theta = \frac{1}{2}[\pi - (A + \delta)]$$

or $$\sin\theta = \frac{b}{a} = \cos\frac{A+\delta}{2}$$

where the length a is shown in the figure. Further

$$\sin\frac{A}{2} = \frac{t/2}{a}$$

where t is the length of the base of the prism. Thus,

$$\Delta\delta \approx \frac{t}{b}\frac{dn}{d\lambda}\Delta\lambda \qquad \textbf{(18.72)}$$

Substituting in Eq. (18.71) we get for the resolving power

$$R = \frac{\lambda}{\Delta\lambda} = t\frac{dn}{d\lambda} \qquad \textbf{(18.73)}$$

Now, for most glasses, the wavelength dependence of the refractive index (in the visible region of the spectrum) can be accurately described by the Cauchy formula

$$n = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} + \ldots \qquad \textbf{(18.74)}$$

Thus,

$$\frac{dn}{d\lambda} = -\left[\frac{2B}{\lambda^3} + \frac{4C}{\lambda^5} + \ldots\right] \qquad \textbf{(18.75)}$$

the negative sign implying that the refractive index decreases with increase in wavelength. As an example, we consider telescope crown glass for which*

$A = 1.51375$, $B = 4.608 \times 10^{-11}$ cm^2, $C = 6.88 \times 10^{-22}$ cm^4

For $\lambda = 6 \times 10^{-5}$ cm, we have

$$\frac{dn}{d\lambda} \simeq -[4.27 \times 10^2 + 3.54]$$
$$\simeq -4.30 \times 10^2 \text{ cm}^{-1}$$

Thus, for $t \simeq 2.5$ cm we have

$$R = \frac{\lambda}{\Delta\lambda} \simeq 1000$$

which is an order of magnitude less than for typical diffraction gratings with 15,000 lines.

18.9 OBLIQUE INCIDENCE LO 9

Till now we have assumed plane waves incident normally on the grating. For experimental setting, it is quite difficult to achieve the condition of normal incidence to a great precision and it is easily seen that slight deviations from normal incidence will introduce considerable errors. It is, therefore, more practical to consider the more general oblique incidence case (see Fig. 18.41). The wavelength measurement can be carried out by using the method of minimum deviation as we do for prisms.

If the angle of incidence is i, then the path difference of the diffracted rays from two corresponding points in adjacent slits will be $d \sin\theta + d \sin i$ (see Fig. 18.41). Thus, principal maxima will occur when

$$d(\sin\theta + \sin i) = m\lambda \qquad \textbf{(18.76)}$$

or $$d[\sin(\delta - i) + \sin i] = m\lambda \qquad \textbf{(18.77)}$$

when $\delta = i + \theta$ is the angle of deviation. For δ to be minimum we must have

$$\frac{d}{di}[\sin(\delta - i) + \sin i] = 0 \qquad \textbf{(18.78)}$$

$$-\cos(\delta - i) + \cos i = 0$$

i.e., $$i = \delta - i = \theta \qquad \textbf{(18.79)}$$

or

$$i = \frac{\delta}{2} = \theta \qquad \textbf{(18.80)}$$

Hence, at the position of minimum deviation, the grating condition becomes

$$2d\sin\frac{\delta}{2} = m\lambda \qquad \textbf{(18.81)}$$

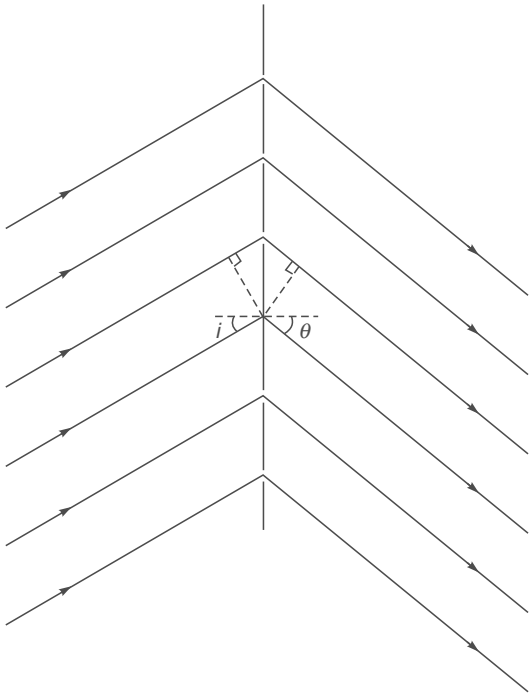


Fig. 18.41 Diffraction of a plane wave incident obliquely on a grating.

* Data quoted from Ref. 18.2.

The minimum deviation position can be obtained in a manner similar to that used in the case of a prism and since the adjustments are relatively simpler, this provides a more accurate method for the determination of λ.

18.10 X-RAY DIFFRACTION* LO 9

Visible light is an electromagnetic wave whose wavelength approximately lies between 4000 Å and 7000 Å. X-rays are also electromagnetic waves whose wavelengths are ~ 1 Å. Obviously, it is extremely difficult to make slits which are narrow enough for the study of X-ray diffraction patterns. Since the interatomic spacings in a crystal are usually of the order of Angstroms, one can use it as a three-dimensional diffraction grating for studying the diffraction of X-rays. Indeed, X-rays have extensively been used to study crystal structures[7].

In an ideal crystal, the atoms or molecules arrange themselves in a regular three-dimensional pattern which can be obtained by a three-dimensional repetition of a certain unit pattern. This simplest volume which has all the characteristics of the whole crystal and which completely fills space is called the unit cell. One can think of various identifiable planes in the regular three-dimensional periodic arrangement (see Fig. 18.42). Miller indices are universally used as a system of notation for planes within a crystal. They specify the orientation of planes relative to the crystal axis without giving the position of the plane in space with respect to the origin. These indices are based on the intercepts of a plane with the three crystal axes, each intercept with an axis being measured in terms of unit cell dimensions (a, b or c) along that axis. To determine the Miller indices of a plane, the following procedure is used:

(a) Find the intercepts (of the plane nearest to the origin) on the three axes and express them as multiple or fractions of the unit cell dimension.
(b) Take the reciprocals of these numbers and multiply by the LCM of the denominators.
(c) Enclose in parentheses.

For example, a (111) plane intercepts all three axes at one unit distance (see Fig. 18.43(a); a (211) plane intercepts the three axes at $\frac{1}{2}$, 1 and 1 unit distances (see Fig. 18.43(b)). Similarly, a (110) plane intercepts the z-axis at ∞. Miller indices can also be negative, the minus sign is shown above the digit like $(\bar{1}1\bar{1})$. Figure 18.44 shows the planes characterized by the Miller indices $(\bar{1}11)$ in a simple cubic lattice.

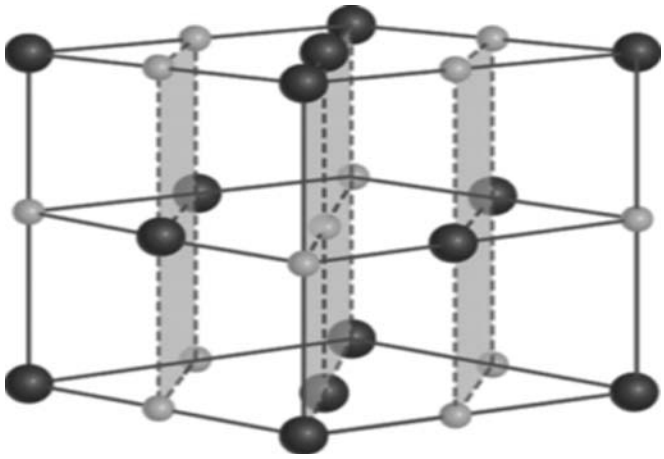


Fig. 18.42 Planes in a NaCl crystal.

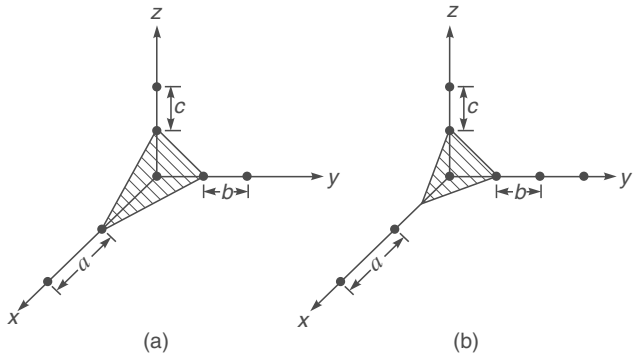


Fig. 18.43 (a) The (111) plane intercepts all three axes at one unit distance of each axial dimension. (b) The (211) plane intercepts the three axes at ½, 1 and 1 unit distances.

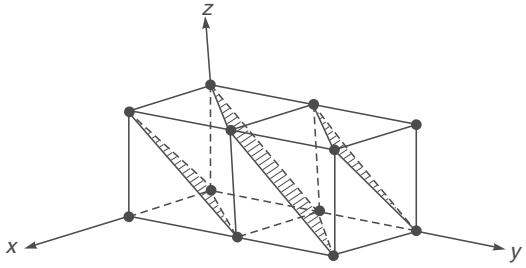


Fig. 18.44 Planes characterized by the Miller indices $(\bar{1}11)$ in a simple cubic lattice.

Consider a monochromatic beam of X-rays to be incident on a crystal. In Fig. 18.45 the horizontal dotted lines represent a set of parallel crystal planes with Miller indices (hkl). W_1 W_2 and W_3W_4 represent the incident and reflected

* The author is grateful to Professor Lalit K. Malhotra for his help in writing this section.

wavefronts, respectively. Obviously, the secondary wavelets emanating from the points A, B and C are in phase on W_3W_4 (see Sec. 12.4 and Fig. 12.7); and the waves emanating from the points A_1, B_1 and C_1 will also be in phase on W_3 W_4 if

$$XB_1 + B_1Y = m\lambda, m = 1, 2, 3,\ldots \quad \textbf{(18.82)}$$

or when

$$2d_{hkl} \sin \theta = m\lambda \quad \textbf{(18.83)}$$

where d_{hkl} is the interplanar spacing between crystal planes of indices (hkl), $m = 1, 2, 3,\ldots$ is called the **order of diffraction** and θ is known as the **glancing angle**. This equation is known as Bragg's law and gives the angular positions of the reinforced diffracted beams in terms of the wavelength λ of the incoming X-rays and of the interplanar spacings d_{hkl} of the crystal planes. When the condition expressed by Eq. (18.83) is not satisfied, destructive interference occurs and no reinforced beam will be produced. Constructive interference occurs when the condition given by Eq. (18.83) is satisfied leading to peaks in the intensity distribution. For solids which crystallize in cubic structures (which are discussed later), the interplanar spacing d_{hkl} between two closest parallel planes with Miller indices (hkl) is given by

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad \textbf{(18.84)}$$

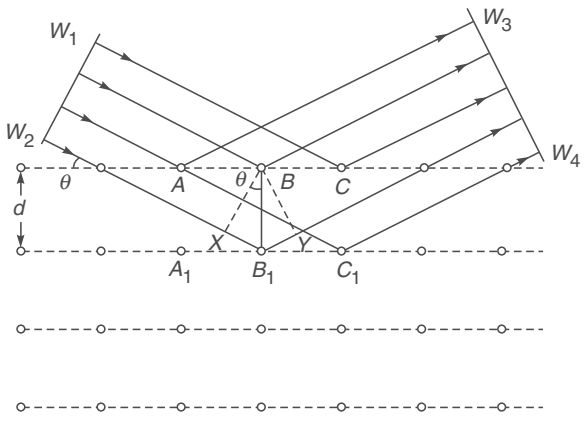


Fig. 18.45 Reflection of a plane wave by a set of parallel crystal planes characterized by the Miller indices (hkl). When the Bragg condition $2d \sin \theta = m\lambda$ is satisfied, the waves scattered from different rows will be in phase.

where a represents the lattice constant. Thus knowing the Miller indices, we can find d_{hkl} and from Bragg's law, we can determine the value of θ at which Bragg's equation can be satisfied.

There are three types of cubic structures: simple cubic, body centred cubic (BCC) and face centred cubic (FCC). Figure 18.45 shows a simple cubic structure (abbreviated as SC) in which the atoms are at the corners of a cube which forms what is known as a **unit cell**. The crystal is built up by the repetition of this unit cell in three dimensions. In addition, if there is an atom at the centre of each cube (shown as 9, 10, 11 and 12 in Fig. 18.46), the arrangement is known as a **BCC structure**. The distance between two adjacent planes characterized by the Miller indices $(\bar{1}10)$ is $a/\sqrt{2}$ which can be verified by simple geometry. On the other hand, if instead of having an atom at the center of the cube there is an atom at the center of each of the six faces of the cube (see Fig. 18.47) we will have the FCC structure. Copper, silver and gold crystallize in the FCC form with the lattice parameter a = 3.61 Å, 4.09 Å and 4.08 Å, respectively. Metals like sodium, barium and tungsten crystallize in the BCC form with a = 4.29 Å, 5.03 Å and 3.16 Å, respectively.

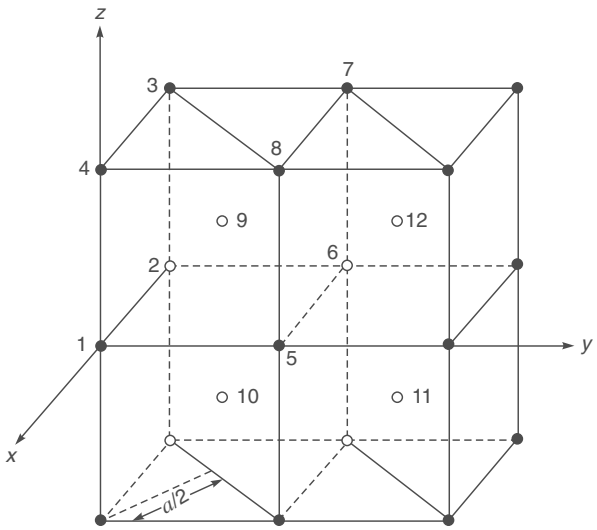


Fig. 18.46 A body centered cubic (bcc) lattice. The $(\bar{1}10)$ planes are separated by $a/\sqrt{2}$.

Just as there are optical missing orders of a diffraction grating, there are structural extinctions of X-ray reflection from a crystal. For simple cubic structures, reflections from all (hkl) planes are possible. However for the BCC structure, diffraction occurs only on planes whose Miller indices when added together total to an even number. Thus for the BCC structure, the principal diffracting planes for a first order diffraction are (110), (200), (211) (and other similar planes), etc. where $h + k + l$ is an even number. In the case of the FCC crystal structure, the principal diffracting planes are those whose Miller indices are either all even or all odd, e.g., (111), (200), (220), etc.

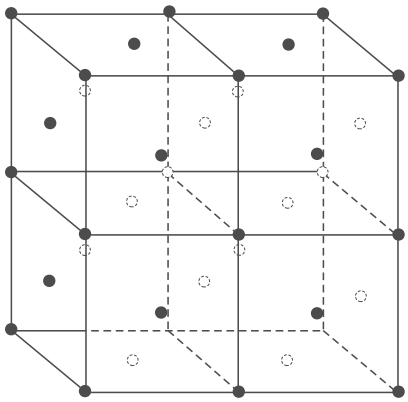


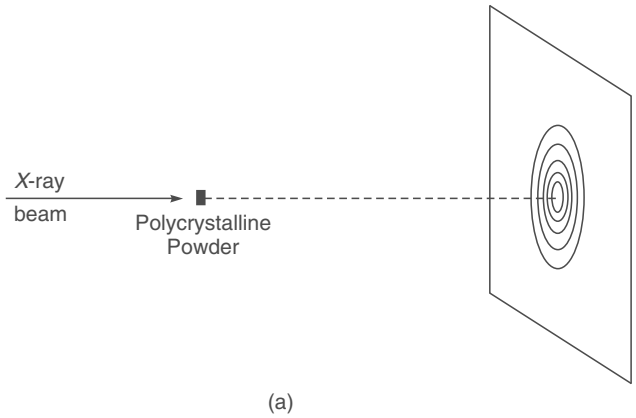
Fig. 18.47 A face centered cubic (fcc) lattice.

18.10.1 Experimental Methods of X-ray Diffraction

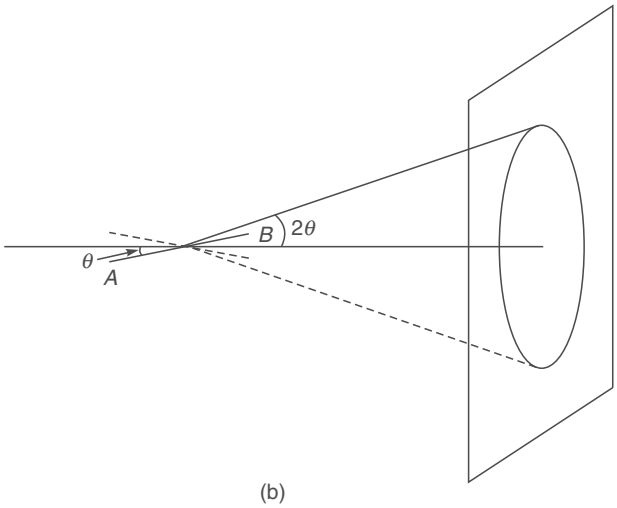
From Bragg's law, $2d_{hkl} \sin\theta = m\lambda$, it is clear that there are essentially three methods which can be used so that Bragg's formula can be satisfied:

	λ	θ
Rotating crystal method	Fixed	Variable (intentional)
Powder method	Fixed	Variable (inherent)
Laue method	Variable	Fixed

When one uses monochomatic X-rays, Bragg's formula cannot be satisfied for an arbitrary value of θ. Hence one rotates the single crystal so that reflection can occur for a discrete set of θ values. This method can only be employed if single crystals of reasonable size are available. If this is not the case, one can still use monochromatic X-rays provided the sample is in the powder form so that there are always enough crystallites of the right orientation available to satisfy the Bragg relation. A powder will consist of a large number of randomly oriented micro-crystals; each micro-crystal is essentially a single crystal. As the X-ray beam passes through such a polycrystalline material, the orientation of any given set of planes, with reference to the X-ray beam, changes from one micro-crystal to the other. Thus, corresponding to any given set of planes there will be a large number of crystals for which Bragg's condition will be satisfied, and on the photographic plate one will obtain concentric rings [see Fig. 18.48 (a)]; each ring will correspond to a particular value of d_{hkl} and a particular value of m. The appearance of the circular rings can be understood as follows. Consider a set of planes parallel to AB [see Fig. 18.48 (b)]. The glancing angle θ is assumed to satisfy



(a)



(b)

Fig. 18.48 (a) When a monochromatic X-ray beam falls on a polycrystalline sample one obtains the Debye–Scherrer rings. (b) Diffraction from a polycrystalline sample.

the Bragg condition. If the microcrystal is rotated about the direction of the incident X-ray beam, then for all positions of the microcrystal, the glacing angle will be the same for these sets of planes. Further, for each position of the microcrystal, the direction of the diffracted beam will be different, but it will always lie on the surface of the cone whose semi-vertical angle will be 2θ. Consequently, one will obtain concentric circular rings on the photographic plate; these rings are known as **Debye–Scherrer rings**.

While using the powder method, the photographic film is put in a cylindrical form surrounding the polycrystalline sample as shown in Fig. 18.49 (a). Each Debye–Scherrer ring will produce an arc on the film, and when the film is unrolled, one obtains a pattern as shown in Figs. 18.49 (b) and (c). From the position of these arcs, one can calculate θ and thus determine the interplanar spacing. From a study of the

interplanar spacings one can determine the crystal structure*. Although a powder camera with an enclosed film strip has been extensively used in the past, modern X-ray crystal analysis uses an X-ray diffractometer which has a radiation counter to detect the angle and intensity of the diffracted beam.

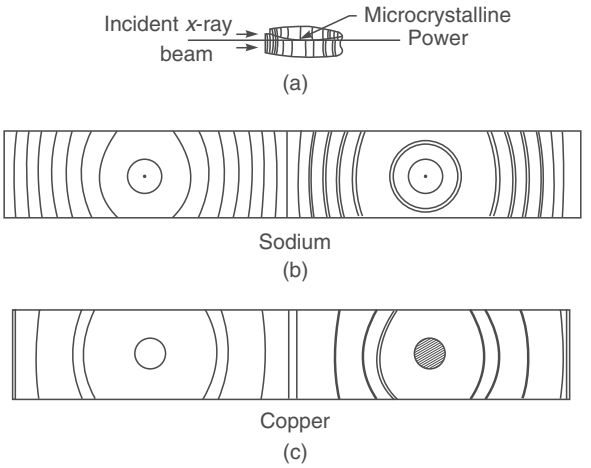


Fig. 18.49 (a) While using the powder method the photographic film is kept in a cylindrical form as shown in the figure. (b) and (c) represent schematic diffraction patterns for sodium and copper, respectively.

Finally there is the Laue method in which the single crystal is held stationary in a beam of white X-rays. Each set of planes then chooses its own wavelength to satisfy the Bragg relation (see Fig. 18.50).

In order to calculate the angles of diffraction we substitute Eq. (18.84) in the Bragg's law [Eq. (18.83)] to obtain

$$\frac{2a}{\sqrt{h^2 + k^2 + l^2}} \sin\theta = m\lambda \qquad \textbf{(18.85)}$$

We restrict ourselves only to first order reflections (m = 1); higher order reflections are usually rather weak (see also Problem 18.22). Thus Eq. (18.85) can be written in the form:

$$\sin\theta = \frac{\lambda}{2a}\sqrt{N} \qquad \textbf{(18.86)}$$

where

$$N = h^2 + k^2 + l^2$$

Now, for a simple cubic lattice, all values of (hkl) are possible implying the following possible values of N:

$$N = 1, 2, 3, 4, 5, 6, 7, \ldots \text{ (SC)} \qquad \textbf{(18.87a)}$$

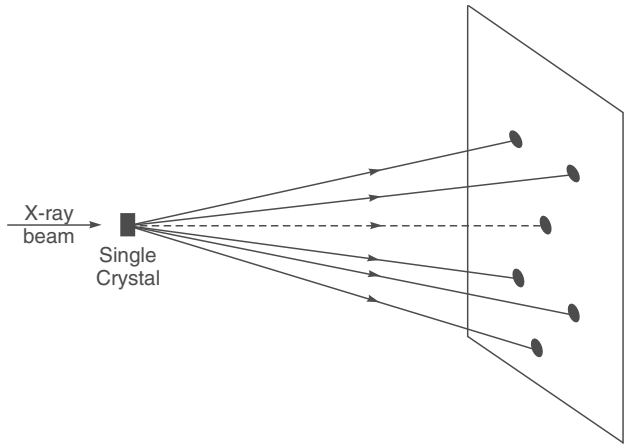


Fig. 18.50 When a polychromatic X-ray beam falls on a single crystal, one obtains Laue spots. Each set of planes chooses its own wavelength to satisfy the Bragg relation given by Eq. (18.84); see also Fig. 25 in the prelim pages.

Similarly, for a BCC lattice $h + k + l$ must be even implying

$$N = h^2 + k^2 + l^2$$
$$= 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, \ldots \text{(BCC)} \qquad \textbf{(18.87b)}$$

Finally, for an FCC lattice, Miller indices are either all even or all odd implying

$$N = h^2 + k^2 + l^2$$
$$= 3, 4, 8, 11, 12, 16, 19, 20, 24, 27, \ldots \text{ (FCC)} \qquad \textbf{(18.87c)}$$

For a given structure and for given values of λ and a one can now easily calculate the different values of θ. For example, if we consider λ = 1.540 Å and 1.544 Å (corresponding to the $CuK_{\alpha 1}$ and $CuK_{\alpha 2}$ lines) then for sodium (which is a BCC structure with a = 4.2906 Å), the various values of θ are

(14.70°, 14.74°),	(21.03°, 21.09°),	(26.08°, 26.15°),
(30.50°, 30.59°),	(34.58°, 34.68°),	(38.44°, 38.56°),
(42.18°, 42.32°),	(45.88°, 46.03°),	(49.59°, 49.76°),
(53.38°, 53.58°),	(57.33°, 57.56°),	(61.54°, 61.82°),
(66.22°, 66.56°),	(79.41°, 80.23°),	

The two values inside the parentheses correspond to the two wavelengths 1.540 Å and 1.544 Å, respectively. Because of the presence of two wavelengths one obtains double lines for each family of planes which become resolvable only at higher scattering angles. Similarly one can consider reflections from other structures (see Problems 18.19, 18.20 and

* Crystal structures other than cubic are also common; for example, zinc crystallizes into a hexagonal structure and carbon forms a diamond structure. However, the most important fact is that in all these structures there is a definite periodicity of atoms.

18.21). Each value of θ will give rise to a Debye–Scherrer ring shown in Figs. 18.48 (a), 18.49 (b) and 18.49 (c).

Finally, we should mention that the intensity of the diffracted wave depends on the number of atoms per unit area in the plane under consideration. For example, corresponding to the $(\bar{1}10)$ and the $(\bar{2}22)$ planes passing through a BCC lattice, there will be one atom and two atoms, respectively, in an area a^2. Thus in the first case the intensity of the diffracted wave will be much more than in the second case.

18.11 THE SELF-FOCUSING PHENOMENON* LO 10

With the availability of intense laser beams, a large number of interesting non-linear optical phenomena have been investigated. One such non-linear phenomenon is the effect on the propagation of a light beam due to the dependence of the refractive index on the intensity of the beam. This leads to the self-focusing (or defocusing) of the beam. In order to physically understand the self-focusing phenomenon we assume the non-linear dependence of the refractive index on the intensity to be of the form

$$n = n_0 + \frac{1}{2} n' E_0^2 \qquad \textbf{(18.88)}$$

where n_0 is the refractive index of the medium in the absence of the electromagnetic field, n' is a constant representing the non-linear effect** and E_0 representing the amplitude of the electric field. As an example, we consider the incidence of a laser beam (propagating in the z-direction) having Gaussian intensity distribution in the transverse direction, i.e., we assume

$$E(x, y, z, t) \approx E_0 \cos(kz - \omega t) \qquad \textbf{(18.89)}$$

with

$$E_0 = E_{00} \exp\left(-\frac{r^2}{a^2}\right) \qquad \textbf{(18.90)}$$

where a represents the width of the Gaussian beam and r ($= \sqrt{x^2 + y^2}$) represents the cylindrical coordinate. In the absence of any nonlinear effects the beam will undergo diffraction divergence (see Sec. 20.5). However, if the beam is incident on a medium characterized by a positive value of n', the intensity distribution will create a refractive index distribution which will have a maximum value on the axis (i.e., at $r = 0$) and will gradually decrease with r. Indeed, using Eqs. (18.88)–(18.90) we will have

$$n \simeq n_0 + \frac{1}{2} n' E_{00}^2 \exp\left(-\frac{2r^2}{a^2}\right)$$

$$\simeq \left(n_0 + \frac{1}{2} n' E_{00}^2\right) - \frac{1}{2} n_0 \left(\frac{r}{\alpha}\right)^2 \qquad \textbf{(18.91)}$$

where

$$\alpha^2 = \frac{n_0 a^2}{2n' E_{00}^2} \qquad \textbf{(18.92)}$$

and in writing Eq. (18.91) we have expanded the exponential term and have retained only the first two terms. In other words, we are restricting ourselves to small values of r, which is the paraxial approximation. The term $\frac{1}{2} n' E_{00}^2$ is usually very small compared to n_0; so we may write (after squaring)

$$n^2 \simeq n_0^2 \left[1 - \left(\frac{r}{\alpha}\right)^2\right] \qquad \textbf{(18.93)}$$

We may recall that in Sec. 3.4.1 we had considered propagation in a medium whose refractive index decreased parabolically from the axis and had shown that the beam could undergo periodic focusing (see Fig. 3.25). Indeed we had shown that the medium behaved like a converging lens of focal length $\pi\alpha/2$ [see Eq. (3.48) of Chapter 3]. In the present case also because of nonlinear effects (with $n' > 0$), the medium will act as a converging lens of focal length approximately given by

* Based on Ref. 18.8; for a rigorous account, e.g., Ref. 18.9.

** This dependence may arise from a variety of mechanisms, such as the Kerr effect, electrostriction, thermal effect, etc. The simplest to understand is the thermal effect which is due to the fact that when an intense optical beam having a transverse distribution of intensity propagates through an absorbing medium, a temperature gradient is set up. For example, if the beam has a Gaussian transverse intensity variation (i.e., of the form $\exp(-r^2/a^2)$; the direction of propagation being along the z-axis), then the temperature will be maximum on the axis (i.e., $r = 0$) and will decrease with increase in the value of r. If $dn/dT > 0$, the refractive index will be maximum on the axis and the beam will undergo focusing; on the other hand if $dn/dT < 0$, the beam will undergo defocusing (see, e.g., Ref. 18.9).

The Kerr effect arises due to the anisotropic polarizability of liquid molecules (like CS_2). An intense light wave will tend to orient the anisotropically polarized molecules such that the direction of maximum polarizability is along the direction of the electric vector; this changes the dielectric constant of the medium. On the other hand, electrostriction (which is important in solids) is the force which a non-uniform electric field exerts on a material medium; this force affects the density of the material, which in turn affects the refractive index. Thus, a beam having nonuniform intensity distribution along its wavefront will give rise to a refractive index variation leading to the focusing (or defocusing) of the beam. For a detailed discussion on electrostriction and Kerr effect, refer to Refs. 18.9–18.11.

$$f_{nl} \simeq \frac{\pi}{2}\alpha \simeq \frac{\pi}{2}\left[\frac{n_0}{2n' E_{00}^2}\right]^{1/2} a \qquad (18.94)$$

the subscript (nl) signifying that the effect is due to a non-linear phenomenon. Thus, because of non-linear effects the beam is said to undergo *self-focusing;* the word *self* signifies the fact that the beam creates its own refractive index gradient resulting in the focusing of the beam*.

Our analysis in Sec. 3.4.1 for the calculation of the focal length was based on ray optics and neglected diffraction effects. Now, in the absence of any non-linear effects, the beam will spread out due to diffraction and the angle of divergence will approximately be given by (see Fig. 18.14)

$$\theta_d \simeq \frac{\lambda}{\pi a} = \frac{(\lambda_0/n_0)}{\pi a} \qquad (18.95)$$

where λ_0 is the free space wavelength. Thus the phenomenon of diffraction can be approximated by a diverging lens of focal length (see Fig. 18.51).

$$f_d \simeq \frac{a}{\theta_d} \simeq \frac{1}{2}ka^2 \qquad (18.96)$$

where $$k = \frac{2\pi}{\lambda} = \frac{2\pi}{\lambda_0} n_0 \qquad (18.97)$$

Clearly if $f_d < f_{nl}$, the diffraction divergence will dominate and the beam will diverge. On the other hand, if $f_{nl} < f_d$, the non-linear focusing effects will dominate and the beam will undergo self-focusing. For $f_d \approx f_{nl}$, the two effects will cancel each other and the beam will propagate without any focusing or defocusing. This is the condition of *uniform waveguide like propagation*. In order to determine the critical power of the beam we note that the condition $f_d \approx f_{nl}$ implies

$$\frac{1}{2}ka^2 \approx \frac{\pi}{2}\left[\frac{n_0}{2n' E_{00}^2}\right]^{1/2} a$$

or

$$E_{00}^2 \simeq \frac{1}{n_0 n'}\frac{\lambda_0^2}{8a^2} \qquad (18.98)$$

Now the total power of the beam is given by

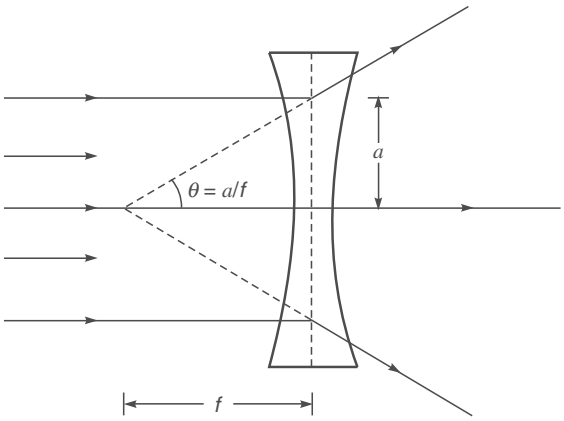


Fig. 18.51 When a plane wave is incident on a diverging lens, the transmitted rays diverge making an angle $\theta \approx a/f$ with the axis.

$$P = \int_0^\infty \text{velocity} \times (\text{energy/unit volume}) \times 2\pi r\, dr$$

$$\simeq \int_0^\infty \left(\frac{c}{n_0}\right) \times \left(\frac{1}{2}\varepsilon E_0^2\right) \times 2\pi r\, dr$$

$$\simeq \frac{c}{n_0}\left(\frac{1}{2}n_0^2 \varepsilon_0 E_{00}^2\right) \int_0^\infty \exp\left(-\frac{2r^2}{a^2}\right) 2\pi r\, dr$$

$$\simeq \frac{\pi}{4} n_0 c \varepsilon_0 E_{00}^2 a^2 \qquad (18.99)$$

where ε ($= n_{00}^2 \varepsilon_0$) is the dielectric permittivity of the medium and ε_0 ($= 8.85 \times 10^{-12}$ C^2/N–m^2) is the dielectric permittivity of free space (see Sec. 19.2). Substituting the expression for E_{00}^2 from Eq. (18.98) in Eq. (18.99) we obtain the following expression for the critical power:

$$P_{cr} \simeq \frac{\pi}{32}(c\varepsilon_0)\frac{\lambda_0^2}{n'} \qquad (18.100)$$

Garmire, Chiao and Townes (Ref. 18.12) carried out experiments on the self-focusing of a ruby laser beam ($\lambda_0 = 0.6943\ \mu$m) in CS_2 and found that the critical power was 25 ± 5 kW. Equation (18.100) gives us

$$P_{cr} \simeq \frac{3.14}{32} \times 3 \times 10^8 \times 8.85 \times 10^{-12} \times \frac{(0.6943 \times 10^{-6})^2}{2 \times 10^{-20}}$$

$$\simeq 6.3 \text{ kW} \qquad (18.101)$$

* It should be mentioned that if n' were a negative quantity, the refractive index would have increased as we move away from the axis and the beam would have undergone defocusing. For example, if the refractive index decreases with increase in temperature, the beam may undergo what is known as *thermal defocusing*.

where we have used the following parameters for CS_2: $n_0 \simeq 1.6276$, $n' \simeq 1.8 \times 10^{-11}$ cgs units $\simeq 2 \times 10^{-20}$ mks units. [The mks unit for n' is (meter/volt)^2.] Although the result is wrong by a factor of about 4, one does obtain the correct order; this is indeed the case for all order-of-magnitude calculations. Thus,

(a) when $P < P_{cr}$, the beam will diverge due to diffraction.
(b) when $P = P_{cr}$, the beam will propagate without divergence or convergence. This is the condition for uniform waveguide propagation.
(c) when $P > P_{cr}$, we may extrapolate that the beam will undergo focusing, which is indeed borne out by more rigorous analysis. This is known as the **self-focusing of the beam**.

We may mention that a detailed study of the self-focusing phenomenon is of considerable importance in laser induced fusion experiments where there is a non-linear interaction of the laser beam with the plasma.

Summary

- Interference corresponds to the situation when we consider the superposition of waves coming out from a number of point sources and diffraction corresponds to the situation when we consider waves coming out from an area source like a circular or rectangular aperture or even a large number of rectangular apertures (like the diffraction grating).
- When a plane wave is incident normally on N parallel slits, the Fraunhofer diffraction pattern is given by

$$I = I_0 \frac{\sin^2 \beta}{\beta^2} \cdot \frac{\sin^2 N\gamma}{\sin^2 \gamma}$$

where

$$\beta = \frac{\pi b \sin\theta}{\lambda}, \quad \gamma = \frac{\pi d \sin\theta}{\lambda}$$

λ is the wavelength of light, θ the angle of diffraction, b represents the width of each slit and d the separation between two slits. When $N = 1$, we have the single slit diffraction pattern producing a central maximum at $\theta = 0$ and minima when $b \sin\theta = m\lambda$; $m = \pm 1, \pm 2, \ldots$ When $N \geq 2$, the intensity distribution is the product of the single slit diffraction pattern and the interference pattern produced by N point sources separated by a distance d. For $N = 2$, we obtain the Young's double slit interference pattern. For large values of N, the principle maxima occur when $\gamma = m\pi$ implying

$$d \sin\theta = m\lambda; \; m = 0, 1, 2, \ldots$$

which is usually referred to as the grating condition.

- The resolving power of the grating is given by

$$R = \frac{\lambda}{\Delta\lambda} = mN$$

where N represents the total number of lines in the grating. For example, in the first order spectrum ($m = 1$) of a diffraction grating with $N = 10000$, for $\lambda \approx 5000$ Å we get $\Delta\lambda \approx 0.5$ Å.

- Consider a monochromatic beam of X–rays incident on a crystal. The glancing angle θ for which we have reinforced diffracted beams is given by

$$2d_{hkl} \sin\theta = m\lambda$$

where d_{hkl} is the interplanar spacing between crystal planes having Miller indices (hkl); $m = 1, 2, 3, \ldots$ is called the order of diffraction and θ is known as the glancing angle. The above equation is known as Bragg's law and gives the angular positions of the reinforced diffracted beams.

Problems

18.1 A plane wave ($\lambda = 5000$ Å) falls normally on a long narrow slit of width 0.5 mm. Calculate the angles of diffraction corresponding to the first three minima. Repeat the calculations corresponding to a slit width of 0.1 mm. Interpret physically the change in the diffraction pattern.

[**Ans:** 0.057°, 0.115°, 0.17°; 0.29°, 0.57°, 0.86°]

18.2 A convex lens of focal length 20 cm is placed after a slit of width 0.6 mm. If a plane wave of wavelength 6000 Å falls normally on the slit, calculate the separation between the second minima on either side of the central maximum.

[**Ans:** $\simeq$ 0.08 cm]

18.3 In Problem 18.2 calculate the ratio of the intensity of the principal maximum to the first maximum on either side of the principal maximum.

[**Ans:** ~ 21]

18.4 Consider a laser beam of circular crosssection of diameter 3 cm and of wavelength 5×10^{-5} cm. Calculate the order of the beam diameter after it has traversed a distance of 3 km.

[**Ans:** ~ 14 cm. This shows the extremely high directionality of laser beams.]

18.5 A circular aperture of radius 0.01 cm is placed in front of a convex lens of focal length 25 cm and illuminated by a parallel beam of light of wavelength 5×10^{-5} cm. Calculate the radii of the first three dark rings.

[**Ans:** 0.76, 1.4, 2.02 mm]

18.6 Consider a plane wave incident on a convex lens of diameter 5 cm and of focal length 10 cm. If the wavelength of the incident light is 6000 Å, calculate the radius of the first dark ring on the focal plane of the lens. Repeat the calculations for a lens of same focal length but diameter 15 cm. Interpret the results physically.

[**Ans:** 1.46×10^{-4} cm, 4.88×10^{-5} cm]

18.7 Consider a set of two slits each of width $b = 5 \times 10^{-2}$ cm and separated by a distance $d = 0.1$ cm, illuminated by a monochromatic light of wavelength 6.328×10^{-5} cm. If a convex lens of focal length 10 cm is placed beyond the

double slit arrangement, calculate the positions of the minima inside the first diffraction minimum.

[**Ans:** 0.0316 mm, 0.094 mm]

18.8 Show that when $b = d$, the resulting diffraction pattern corresponds to a slit of width $2b$.

18.9 Show that the first order and second order spectra will never overlap when the grating is used for studying a light beam containing wavelength components from 4000 Å to 7000 Å.

18.10 Consider a diffraction grating of width 5 cm with slits of width 0.0001 cm separated by a distance of 0.0002 cm. What is the corresponding grating element? How many order would be observable at $\lambda = 5.5 \times 10^{-5}$ cm? Calculate the width of the principal maximum. Would there be any missing orders?

18.11 For the diffraction grating of Problem 18.10, calculate the dispersion in the different orders. What will be the resolving power in each order?

18.12 A grating (with 15000 lines per inch) is illuminated by white light. Assuming that white light consists of wavelengths lying between 4000 and 7000 Å, calculate the angular widths of the first and the second order spectra. [*Hint: You should not use Eq. (18.65); why*?]

18.13 A grating (with 15,000 lines per inch) is illuminated by sodium light. The grating spectrum is observed on the focal plane of a convex lens of focal length 10 cm. Calculate the separation between the D_1 and D_2 lines of sodium. (The wavelengths of the D_1 and D_2 lines are 5890 and 5896 Å, respectively.) [*Hint: You may use Eq. (18.65).*]

18.14 Calculate the resolving power in the second order spectrum of a 1 inch grating having 15,000 lines.

18.15 Consider a wire grating of width 1 cm having 1,000 wires. Calculate the angular width of the second order principal maxima and compare the value with the one corresponding to a grating having 5000 lines in 1 cm. Assume $\lambda = 5 \times 10^{-5}$ cm.

18.16 In the minimum deviation position of a diffraction grating the first order spectrum corresponds to an angular deviation of 30°. If $\lambda = 6 \times 10^{-5}$ cm, calculate the grating element.

18.17 Calculate the diameter of a telescope lens if a resolution of 0.1 seconds of arc is required at $\lambda = 6 \times 10^{-5}$ cm.

18.18 Assuming that the resolving power of the eye is determined by diffraction effects only, calculate the maximum distance at which two objects separated by a distance of 2 m can be resolved by the eye. (Assume pupil diameter to be 2 mm and $\lambda = 6000$ Å.)

18.19 **(a)** A pinhole camera is essentially a rectangular box with a tiny pinhole in front. An inverted image of the object is formed on the rear of the box. Consider a parallel beam of light incident normally on the pinhole. If we neglect diffraction effects then the diameter of the image will increase linearly with the diameter of the pinhole. On the other hand, if we assume Fraunhofer diffraction, then the diameter of the first dark ring will go on increasing as we reduce the diameter of the pinhole. Find the pinhole diameter for which the diameter of the geometrical image is approximately equal to the diameter of the first dark ring in the Airy pattern. Assume $\lambda = 6000$ Å and a separation of 15 cm between the pinhole and the rear of the box.

(b) Figure 18.52 shows the quality of the image formed for various values of the diameter of the pinhole. Discuss qualitatively the fact that the image will get blurred if the diameter of the pinhole is too big or too small.

[**Ans:** (a) 0.47 mm]

18.20 Copper is an FCC structure with lattice constant 3.615 Å. An X-ray powder photograph of copper is taken. The X-ray beam consists of wavelengths 1.540 Å and 1.544 Å. Show that diffraction maxima will be observed at θ = (21.64°, 21.70°), (25.21°, 25.28°), (37.05°, 37.16°), (44.94°, 45.09°), (47.55°, 47.71°), (58.43°, 58.67°), (68.20°, 68.58°), (72.29°, 72.76°).

18.21 Tungsten is a BCC structure with lattice constant 3.1648 Å. Show that in the powder photograph of tungsten (corresponding to an X-ray wavelength of 1.542 Å) one would observe diffraction maxima at θ = 20.15°, 29.17°, 36.64°, 43.56°, 50.39°, 57.55°, 65.74° and 77.03°.

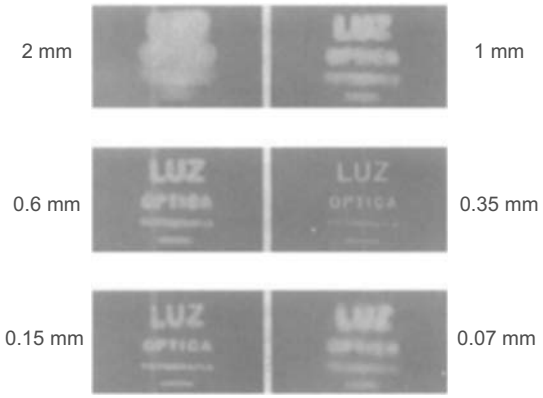


Fig. 18.52 The image formed in a pinhole camera for different diameters of the pinhole. [*Photograph downloaded from the internet by Professor K Thyagarajan; Ref: http://www. cs.berkeley.edu/~daf/book/chapter-4.pdf*].

18.22 **(a)** In the simple cubic structure if we alternately place Na and Cl atoms we would obtain the NaCl structure. Show that the Na atoms (and the Cl atoms) independently form FCC structures. The lattice constant associated with each fcc structure is 5.6402 Å. Corresponding to the X-ray wavelength 1.542 Å, show that diffraction maxima will be observed at θ = 13.69°, 15.86°, 22.75°, 26.95°, 28.27°, 33.15°, 36.57°, 37.69°, 42.05° 45.26°, 50.66°, 53.98°, 55.10°, 59.84°, 63.69°, 65.06°, 71.27°, 77.45° and 80.66°.

(b) Show that if we treat NaCl as a simple cubic structure with lattice parameter 2.82 Å then the maxima at $\theta = 13.69^\circ, 26.95^\circ, 36.57^\circ, 45.26^\circ, 53.98^\circ, 63.69^\circ$ and 77.45° will not be observed. Indeed in the X-ray diffraction pattern of NaCl, the maxima corresponding to these angles will be very weak.

18.23 Show that the mth order reflection from the planes characterized by (hkl) can be considered as the same as the first order reflection from the planes characterized by $(mh\ mk\ ml)$.

18.24 Calculate the Fraunhofer diffraction pattern produced by a double-slit arrangement with slits of widths b and $3b$, with their centres separated by a distance $6b$.

18.25 Consider the propagation of a 1 kW laser beam (λ = 6943 Å, beam diameter $\simeq$ 1 cm) in CS_2. Calculate f_d and f_{nl} and discuss the defocusing (or focusing) of the beam. Repeat the calculations corresponding to a 1000 kW beam and discuss any qualitative differences that exist between the two cases. The data for n_0 and n_2 are given in Sec. 18.10.

18.26 The values of n_0 and n_2 for benzene are 1.5 and 0.6×10^{-10} cgs. units, respectively. Obtain an approximate expression for the critical power.

Chapter Nineteen FRAUNHOFER DIFFRACTION II AND FOURIER OPTICS

Fourier analysis is a ubiquitous tool that has found application to diverse areas of physics and engineering.

—Joseph Goodman in the Preface to *Introduction to Fourier Optics*

Learning Objectives

After reading this chapter, the reader should be able to:

LO 1: derive the Fresnel diffraction integral.
LO 2: understand conditions for Fraunhofer diffraction.
LO 3: describe the Fraunhofer diffraction by a long narrow slit, and by rectangular and circular apertures.
LO 4: discuss Fraunhofer diffraction by an array of identical apertures.
LO 5: explain spatial frequency filtering.
LO 6: demonstrate the Fourier transforming property of a thin lens.

19.1 INTRODUCTION

In this chapter, we will present a more general analysis of the far field diffraction of a plane wave by different types of aperture; this is known as **Fraunhofer diffraction**. We will first derive the formula for what is known as **Fresnel diffraction**, which will be used in the next chapter. We will then make the far field approximation, which will give us the Fraunhofer diffraction pattern; this will be shown to be the Fourier transform of the aperture function. We will also derive the Fourier transforming property of a thin lens that forms the basis of Fourier optics and of spatial frequency filtering.

19.2 THE FRESNEL DIFFRACTION INTEGRAL

LO 1

We consider a plane wave (of amplitude A) incident normally on an aperture as shown in Fig. 19.1. Using Huygens–Fresnel principle, we will calculate the field produced at the point P on a screen SS' which is at a distance z from the aperture. Now, for a spherical wave *diverging* from the origin, the field distribution is given by

$$u \sim \frac{1}{r} e^{ikr}$$

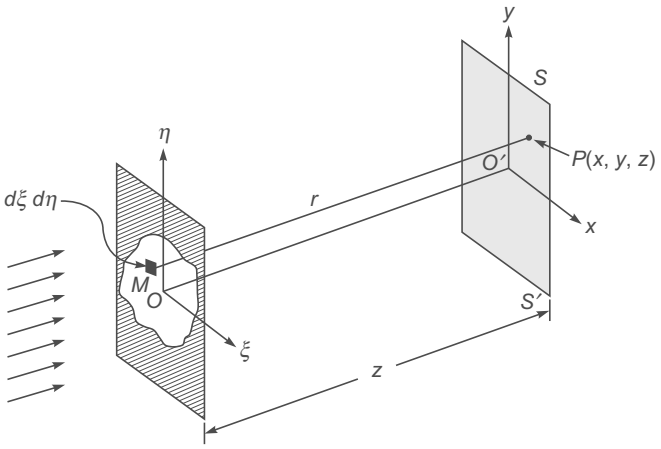


Fig. 19.1 A plane wave incident normally on an aperture. The diffraction pattern is observed on the screen SS'.

where r is the distance from the source (at the origin) to the observation point. We consider an infinitesimal area $d\xi\, d\eta$ (around the point M) on the plane containing the aperture; the field at the point P due to waves emanating from this infinitesimal area will be proportional to

$$\frac{A e^{ikr}}{r} d\xi\, d\eta \qquad \textbf{(19.1)}$$

where $r = MP$. In order to calculate the total field (at the point P), we will have to sum over all the infinitesimal areas (in the aperture) to obtain

$$u(P) = C\iint \frac{A e^{ikr}}{r} d\xi\, d\eta \qquad (19.2)$$

where C is a proportionality constant and the integration is over the entire aperture. From a more general theory, one can show that [see, e.g., Refs. 19.1–19.4]:

$$C = -\frac{ik}{2\pi} = \frac{1}{i\lambda} \qquad (19.3)$$

[see also Sec. 19.3]. We thus obtain

$$u(P = \frac{A}{i\lambda}\iint \frac{e^{ikr}}{r} d\xi\, d\eta \qquad (19.4)$$

If the amplitude and phase distribution on the plane $z = 0$ is given by $A(\xi, \eta)$ then the above integral is modified to

$$u(P) = \frac{1}{i\lambda}\iint A(\xi, \eta)\frac{e^{ikr}}{r} d\xi\, d\eta \qquad (19.5)$$

In writing Eqs. (19.4) and (19.5), we have made two assumptions:

(a) Our analysis has assumed that the screen (in the plane of the aperture) does not affect the field at the point P. This assumption is valid when the dimensions of the aperture are large in comparison to the wavelength. A more accurate analysis would take into consideration the effect of the screen on the field at any point P; this, in general, is a very difficult problem.

(b) We have used a scalar theory in which we have represented the field by a scalar function u; this implies that the electric field is in the same direction everywhere. This assumption will be valid when the line joining the point O and the observation point P makes a small angle with the axis.

The quantity r, which represents the distance between the point M [whose coordinates are $(\xi, \eta, 0)$] on the plane of the aperture and the point P (whose coordinates are x, y, z) on the screen [see Fig. 19.1] will be given by

$$r = [(x-\xi)^2 + (y-\eta)^2 + z^2]^{1/2}$$
$$= z\sqrt{1+\alpha}$$

where

$$\alpha \equiv \frac{(x-\xi)^2}{z^2} + \frac{(y-\eta)^2}{z^2} \qquad (19.6)$$

Now, for $\alpha < 1$, we may write

$$\sqrt{1+\alpha} = 1 + \frac{1}{2}\alpha - \frac{1}{8}\alpha^2 + \ldots \qquad (19.7)$$

If we assume $\alpha << 1$ and neglect quadratic and higher order terms in the above expansion, we would get

$$r \approx z + \frac{(x-\xi)^2}{2z} + \frac{(y-\eta)^2}{2z} \qquad (19.8)$$

Further, in the denominator of Eq. (19.5) we may safely replace r by z so that we may write* (see also Appendix E).

$$u(x, y, z) \approx \frac{1}{i\lambda z} e^{ikz}\iint A(\xi, \eta)\times$$
$$\exp\left\{\frac{ik}{2z}[(x-\xi)^2 + (y-\eta)^2]\right\} d\xi\, d\eta$$

Fresnel Diffraction Integral (19.9)

The above equation can be rewritten in the form

$$u(x, y, z) \approx \frac{1}{i\lambda z} e^{ikz} \exp\left\{\frac{ik}{2z}(x^2 + y^2)\right\}\iint A(\xi, \eta)\times$$
$$\exp\left\{\frac{ik}{2z}(\xi^2 + \eta^2)\right\} e^{-i(u\xi + v\eta)} d\xi\, d\eta \qquad (19.10)$$

where

$$u = \frac{2\pi x}{\lambda z} \quad \text{and} \quad v = \frac{2\pi y}{\lambda z} \qquad (19.11)$$

are known as spatial frequencies. Both Eqs. (19.9) and (19.10) are usually referred to as the **Fresnel diffraction integral**. In the next chapter, we will use the above integrals to calculate the Fresnel diffraction pattern. We must mention here that in the Fresnel approximation, we have neglected the terms proportional to α^2; this will be justified if it leads to maximum phase change which is much less than π. Thus, the Fresnel approximation will be valid when

$$\frac{1}{8} kz\alpha^2 << \pi \Rightarrow \frac{1}{8}\frac{2\pi}{\lambda}\frac{\left[(x-\xi)^2 + (y-\eta)^2\right]^2}{z^3} << \pi \qquad (19.12)$$

* For example, for $\lambda = 6 \times 10^{-5}$ cm, the factor cos kr becomes.

$$\cos\left(\frac{\pi}{3} 10^5 r\right)$$

As the value of r is changed from say 60 cm to 60.00002 cm, the cosine factor will change from +1 to −0.5. This shows the rapidity with which the exponential factor will vary in the domain of integration, although the change in r is extremely small.

Thus, we must have

$$z >> \left(\frac{1}{4\lambda} \left[(x-\xi)^2 + (y-\eta)^2 \right]^2_{\max} \right)^{\frac{1}{3}}$$

Condition for Fresnel Approximation to be valid **(19.13)**

As an example, we consider a circular aperture of radius a; if we observe in a region of dimensions much greater than a, then we may neglect the terms involving ξ and η on the right hand side to obtain:

$$z >> \left(\frac{1}{4\lambda} (x^2 + y^2)^2 \right)^{\frac{1}{3}} \qquad (19.14)$$

Thus for a circular aperture of radius 0.1 cm, if we observe in a radius of about 1 cm the maximum value of $(x^2 + y^2)$ will be about 1 cm^2; if we assume $\lambda \approx 5 \times 10^{-5}$ cm, Eq. (19.14) would imply $z >> 17$ cm.

19.3 UNIFORM AMPLITUDE AND PHASE DISTRIBUTION

We first consider the absence of any aperture. Thus, at $z = 0$

$$A(\xi, \eta) = A \qquad \text{for all values of } \xi \text{ and } \eta$$

and Eq. (19.9) can be written as

$$u(x, y, z) = \frac{A}{i\lambda z} e^{ikz} \int_{-\infty}^{+\infty} e^{\frac{ik}{2z}Y^2} dX \int_{-\infty}^{+\infty} e^{\frac{ik}{2z}Y^2} dY$$

where $X = x - \xi$ and $Y = y - \eta$. If we now use the integral (see Appendix A)

$$\int_{-\infty}^{+\infty} e^{-\alpha x^2 + \beta x} dx = \sqrt{\frac{\pi}{\alpha}} \exp\left[\frac{\beta^2}{4\alpha} \right] \qquad (19.15)$$

we would get

$$u(x, y, z) = \frac{A}{i\lambda z} e^{ikz} \left[\sqrt{\frac{\pi 2z}{-ik}} \right] \left[\sqrt{\frac{\pi 2z}{-ik}} \right]$$

or

$$u(x, y, z) = A e^{ikz} \qquad (19.16)$$

as it indeed should for a uniform plane wave. This shows that in spite of all the approximations that we have made, we have ended up getting the correct result! The above equation also tells us that the value of C given by Eq. (19.3) is correct.

19.4 THE FRAUNHOFER APPROXIMATION LO2

In the Fraunhofer approximation, we assume z to be so large that inside the integral in Eq. (19.10), the function

$$\exp\left\{ \frac{ik}{2z} (\xi^2 + \eta^2) \right\}$$

can be replaced by unity or, the maximum phase change should be much less than π. Thus, **in addition** to the condition given by Eq. (19.13), we must have

$$z >> \frac{[\xi^2 + \eta^2]_{\max}}{\lambda}$$ **Condition for Fraunhofer Approximation to be valid** **(19.17)**

In this approximation, Eq. (19.10) takes the form

$$u(x, y, z) \approx \frac{1}{i\lambda z} e^{ikz} \exp\left\{ \frac{ik}{2z} (x^2 + y^2) \right\} \times$$

$$\iint A(\xi, \eta)\, e^{-i(u\xi + v\eta)} d\xi\, d\eta$$ **Fraunhofer Diffraction Integral** **(19.18)**

which represents the Fraunhofer diffraction pattern. The integral on the right-hand side is the two-dimensional Fourier transform of the function $A(\xi, \eta)$ (see Sec. 9.6). Thus Eq. (19.18) gives the very important result that the

Fraunhofer diffraction pattern is the Fourier transform of the aperture function.

For a circular aperture of radius a, Eq. (19.17) would become

$$z >> \frac{a^2}{\lambda} \qquad (19.19)$$

We introduce the Fresnel number

$$N_F = \frac{a^2}{\lambda z} \qquad (19.20)$$

Thus for the Fraunhofer approximation to be valid, we must have

$$N_F << 1 \qquad (19.21)$$

19.5 FRAUNHOFER DIFFRACTION BY A LONG NARROW SLIT LO 3

We first consider Fraunhofer diffraction of a plane wave incident normally on a long narrow slit of width b (along the ξ-axis) placed on the aperture plane. Figure 19.2 corresponds to a rectangular slit – if the slit is very long along the η-axis, then we will have a long narrow slit. For such a case, we will have

$$A(\xi, \eta) = A \quad |\xi| < \frac{b}{2}$$
$$= 0 \quad |\xi| > \frac{b}{2} \qquad \textbf{(19.22)}$$

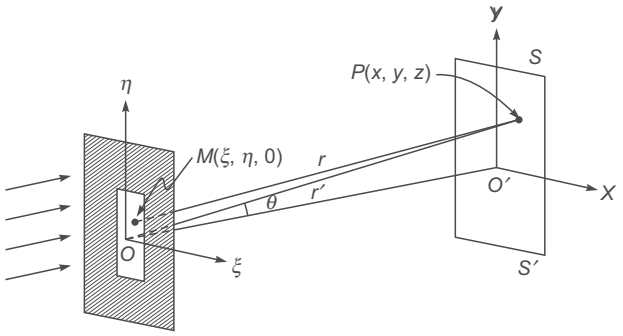


Fig. 19.2 Diffraction of a plane wave incident normally on a rectangular aperture.

for all values of η. Substituting Eq. (19.22) in Eq. (19.18), we obtain

$$u(x, y, z) = \frac{A}{i\lambda z} e^{ikz} \exp\left\{\frac{ik}{2z}(x^2 + y^2)\right\} \times$$
$$\int_{-b/2}^{+b/2} e^{-iu\xi} d\xi \int_{-\infty}^{+\infty} e^{-iv\eta} d\eta \qquad \textbf{(19.23)}$$

Now, in Sec. 9.3 we have shown

$$\delta(v) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-iv\eta} d\eta \qquad \textbf{(19.24)}$$

and

$$\int_{-b/2}^{+b/2} e^{-iu\xi} d\xi = \frac{1}{-iu} e^{-iu\xi}\Big|_{-b/2}^{+b/2}$$
$$= \frac{2}{u} \frac{e^{iub/2} - e^{-iub/2}}{2i} = b \frac{\sin \beta}{\beta}$$

where

$$\beta = \frac{ub}{2} = \frac{\pi b x}{\lambda z} \approx \frac{\pi b \sin \theta}{\lambda} \qquad \textbf{(19.25)}$$

and $\sin \theta \approx \frac{x}{z}$; θ representing the angle of diffraction along the x-direction. Thus,

$$u(x, y, z) = \frac{Ab}{i\lambda z} e^{ikz} \exp\left\{\frac{ik}{2z}(x^2 + y^2)\right\} \left(\frac{\sin \beta}{\beta}\right) 2\pi\delta(v) \qquad \textbf{(19.26)}$$

Because of the δ function, the intensity is zero except on the x-axis; thus the intensity distribution along the x-axis will be

$$I = I_0 \left(\frac{\sin^2 \beta}{\beta^2}\right) \qquad \textbf{(19.27)}$$

We thus obtain the single slit diffraction pattern as discussed in Sec. 18.3. Actually for a long narrow slit, Eq. (19.17) is not valid and we must use the analysis given in Sec. 20.7. However, on the focal plane of a lens we will observe the pattern given by Eq. (19.27) – see Fig. 18.3.

19.6 FRAUNHOFER DIFFRACTION BY A RECTANGULAR APERTURE

LO 3

We next consider a rectangular aperture (of dimension $a \times b$) [see Fig. 19.2]. The Fraunhofer diffraction of a plane wave incident normally on such a rectangular aperture will be given by

$$u(x, y, z) = \frac{A}{i\lambda z} e^{ikz} \exp\left\{\frac{ik}{2z}(x^2 + y^2)\right\}$$
$$\int_{-b/2}^{+b/2} e^{-iu\xi} d\xi \int_{-a/2}^{+a/2} e^{-iv\eta} d\eta \qquad \textbf{(19.28)}$$

where we have chosen the origin to be at the center of the rectangular aperture [see Fig. 19.2]. Carrying out the integration as in the previous section, we obtain

$$u(x, y, z) = \frac{Aba}{i\lambda z} e^{ikz} \exp\left\{\frac{ik}{2z}(x^2 + y^2)\right\} \left(\frac{\sin \beta}{\beta}\right) \left(\frac{\sin \gamma}{\gamma}\right) \qquad \textbf{(19.29)}$$

where β is given by Eq. (19.25),

$$\gamma = \frac{va}{2} = \frac{\pi a y}{\lambda z} \approx \frac{\pi a \sin \phi}{\lambda} \qquad \textbf{(19.30)}$$

and $\sin \phi \approx \frac{y}{z}$; ϕ representing the angle of diffraction along the y-direction. Thus, we may write for the intensity distribution

$$I(P) = I_0 \frac{\sin^2 \gamma}{\gamma^2} \frac{\sin^2 \beta}{\beta^2} \qquad \textbf{(19.31)}$$

The above equation represents the Fraunhofer diffraction pattern by a rectangular aperture. We must remember that Eqs. (19.29) and (19.31) are valid when both Eqs. (19.13) and (19.17) are satisfied. The intensity distribution due to a square aperture ($a = b$) is shown in Fig. 19.3; the figure corresponds to $a = b = 0.01$ cm, $z = 100$ cm and we have assumed $\lambda = 5 \times 10^{-5}$ cm. Now, if we observe in a region of radius 0.5 cm [i.e., $(x^2 + y^2) < 0.25$ cm^2] then Eq. (19.13) would give us

$$z >> \left(\frac{1}{4 \times 5 \times 10^{-5}} (0.25)^2\right)^{\frac{1}{3}} \approx 7 \text{ cm}$$

Further Eq. (19.17) would give us

$$z >> \left(\frac{1}{5 \times 10^{-5}} 2 \times (0.01)^2 \right) \approx 4 \text{ cm}$$

We have chosen $z = 100$ cm and we get the diffraction pattern as shown in Fig. 19.3. Although in the above, we have assumed that we are observing a region of radius 0.5 cm, we have plotted the diffraction pattern for -2 cm $< x, y < +2$ cm. We may note that along the x-axis, the intensity will be zero when

$$\beta = \left(\frac{\pi b x}{\lambda z} \right) = m\pi; \quad m = 0, 1, 2, 3 \ldots \qquad \textbf{(19.32)}$$

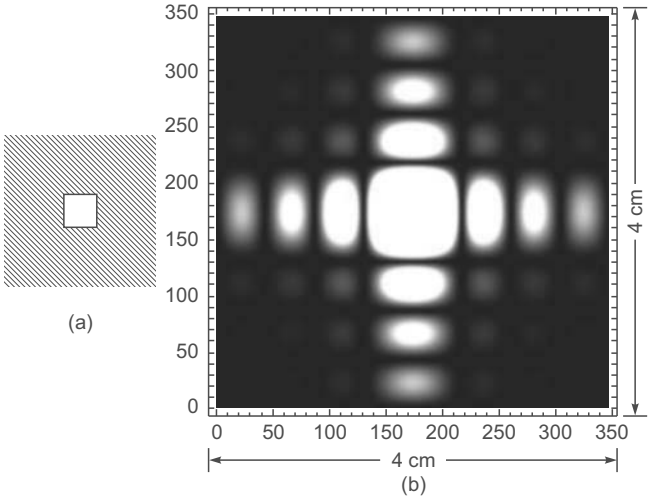


Fig. 19.3 (a) A square aperture of side 0.01 cm. (b) The corresponding (computer generated) Fraunhofer diffraction pattern on a screen at a distance of 100 cm from the aperture; $\lambda = 5 \times 10^{-5}$ cm.

or

$$x = \frac{m\lambda}{b} z = m\, 0.5$$
$$= 0.5 \text{ cm}, 1.0 \text{ cm}, 1.5 \text{ cm}, 2.0 \text{ cm}, \ldots$$

corresponding to $m = 1, 2, 3, 4, \ldots$ respectively; this is consistent with the positions of the minima in Fig. 19.3.

For the case of a long narrow slit (i.e., for $a \to \infty$), the function

$$\frac{a \sin \gamma}{\gamma} = \frac{\sin\left[a \dfrac{\pi \sin \phi}{\lambda} \right]}{\left[\dfrac{\pi \sin \phi}{\lambda} \right]}$$

becomes very sharply peaked around $\phi = 0$. Since $\phi = 0$ implies $y = 0$, there is no diffraction along the y-axis (see Sec. 19.4). The diffraction of a plane wave incident on a long narrow slit and the transition to the Fraunhofer region is discussed in Chapter 20 (see Sec. 20.7)

19.7 FRAUNHOFER DIFFRACTION BY A CIRCULAR APERTURE

LO 3

We consider a plane wave incident normally on a circular aperture as shown in Fig. 19.4. On the plane of the circular aperture we choose cylindrical coordinates [see Fig. 19.5]

$$\xi = \rho \cos \phi \quad \text{and} \quad \eta = \rho \sin \phi \qquad \textbf{(19.33)}$$

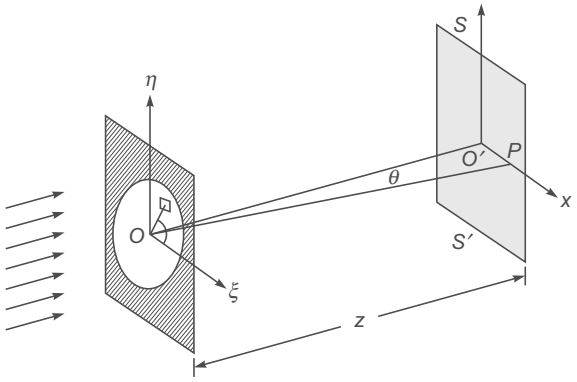


Fig. 19.4 Diffraction of a plane wave incident on a circular aperture of radius a.

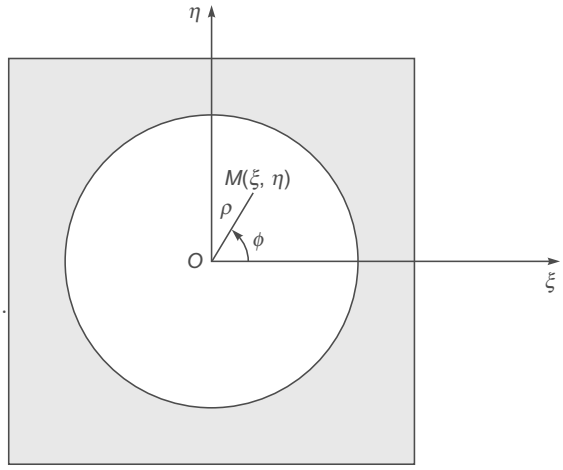


Fig. 19.5 Cylindrical coordinates (ρ, ϕ) on the plane of the circular aperture.

Further, because of the circular symmetry of the system, the diffraction pattern will be of the form of concentric circular rings with their centers at the point O'. Consequently, we may calculate the intensity distribution only along the x-axis (i.e., at points for which $y = 0$) and in the final result replace x by $\sqrt{x^2 + y^2}$. Now, when $y = 0$

$$v = 0 \quad \text{and} \quad \sin \theta \approx \frac{x}{z} \qquad \textbf{(19.34)}$$

where θ is the angle that OP makes with the z-axis. Thus,

$$u = \frac{2\pi x}{\lambda z} = k \sin\theta$$

and Eq. (19.18) becomes

$$u(P) = \frac{A}{i\lambda z} e^{ikz} \exp\left\{\frac{ikr^2}{2z}\right\} \int_0^a \int_0^{2\pi} e^{-ik\rho \sin\theta \cos\phi} \rho \, d\rho \, d\phi \tag{19.35}$$

Thus,

$$u(P) = \frac{A}{i\lambda z} e^{ikz} \exp\left\{\frac{ikr^2}{2z}\right\} \frac{1}{(k \sin\theta)^2} \int_0^{ka\sin\theta} \zeta \, d\zeta \int e^{-i\zeta\cos\phi} \, d\phi$$

$$= \frac{A}{i\lambda z} e^{ikz} \exp\left\{\frac{ikr^2}{2z}\right\} \frac{2\pi}{(k \sin\theta)^2} \int_0^{ka\sin\theta} \zeta J_0(\zeta) d\zeta \tag{19.36}$$

where $\zeta = k\rho \sin\theta$ and use has made of the following well-known relation*

$$J_0(\zeta) = \frac{1}{2\pi} \int_0^{2\pi} e^{\pm i\zeta\cos\phi} \, d\phi \tag{19.37}$$

If we further use the relation

$$\frac{d}{d\zeta}[\zeta J_1(\zeta)] = \zeta J_0(\zeta) \tag{19.38}$$

then Eq. (19.36) becomes

$$u(P) = \frac{A}{i\lambda z} e^{ikz} \exp\left\{\frac{ikr^2}{2z}\right\} \frac{2\pi}{(k \sin\theta)^2} [\zeta J_1(\zeta)]_0^{ka\sin\theta}$$

$$= \frac{A}{i\lambda z} e^{ikz} \exp\left\{\frac{ikr^2}{2z}\right\} \pi a^2 \left[\frac{2 J_1(v)}{v}\right]$$

where $v = k a \sin\theta$. Thus, the intensity distribution would be given by

$$I(P) = I_0 \left[\frac{2 J_1(v)}{v}\right] \tag{19.39}$$

where I_0 is the intensity at the point O' (see Fig. 19.4). This is the famous Airy pattern which has been discussed in Sec. 18.3. We have already mentioned that the diffraction pattern (in the plane SS') will consist of concentric rings with their centers at the point O'. If $F(r)$ represents the fractional energy contained in a circle of radius r, then

$$F(r) = \frac{\int_0^r I(\sigma) 2\pi\sigma \, d\sigma}{\int_0^\infty I(\sigma) 2\pi\sigma \, d\sigma} \tag{19.40}$$

where $I(\sigma)\ 2\pi\sigma\ d\sigma$ would be proportional to the energy contained in the annular region whose radii lie between σ and $\sigma + d\sigma$. Clearly,

$$\sin\theta \approx \frac{\sigma}{z} \tag{19.41}$$

Since $v = k\,a \sin\theta$, we obtain

$$\sigma = \frac{z}{ka} v \tag{19.42}$$

and Eq. (19.40) becomes

$$F(r) = \frac{\int_0^v \left[\frac{2 J_1(v)}{v}\right]^2 v \, dv}{\int_0^\infty \left[\frac{2 J_1(v)}{v}\right]^2 v \, dv} \tag{19.43}$$

where we have used Eq. (19.39) for the intensity distribution. Now,

$$\frac{J_1^2(v)}{v} = J_1(v)\left[J_0(v) - \frac{dJ_1(v)}{dv}\right]$$

$$= -\left[J_0(v)\frac{dJ_0(v)}{dv} + J_1(v)\frac{dJ_1(v)}{dv}\right]$$

$$= -\frac{1}{2}\frac{d}{dv}[J_0^2(v) + J_1^2(v)] \tag{19.44}$$

Thus,

$$F(r) = \frac{J_0^2(v) + J_1^2(v)\Big|_0^v}{J_0^2(v) + J_1^2(v)\Big|_0^\infty} = 1 - J_0^2(v) - J_1^2(v) \tag{19.45}$$

The above function is plotted in Fig. 19.6; one can deduce from the curve that about 84% of light is contained within the circle bounded by the first dark ring and about 91% of the light is contained in the circle bounded by the first two dark rings, etc. The Fruanhofer pattern by an annular aperture is discussed in Problem 19.5.

* The identities associated with Bessel functions can be found in most books on mathematical physics; see, e.g., Ref. 19.5–19.7.

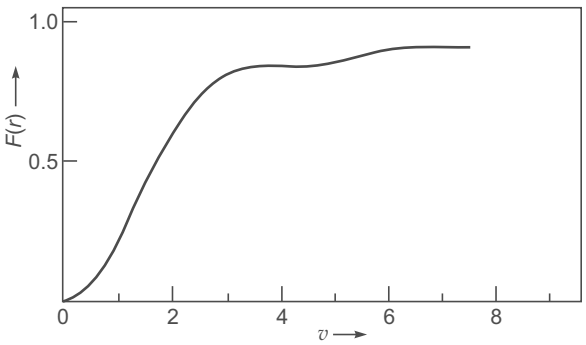


Fig. 19.6 The fractional energy contained in a circle of radius r.

19.8 ARRAY OF IDENTICAL APERTURES

LO 4

We next consider an array of N identical apertures as shown in Fig. 19.7. The Fraunhofer diffraction pattern will be the sum of the fields produced by the individual apertures and will be given by [see Eq. (19.18)]:

$$u = C\left[\iint_{S_1} + \iint_{S_2} + \ldots\right]\exp\left[-i(u\xi + v\eta)\right] d\xi\, d\eta \qquad \textbf{(19.46)}$$

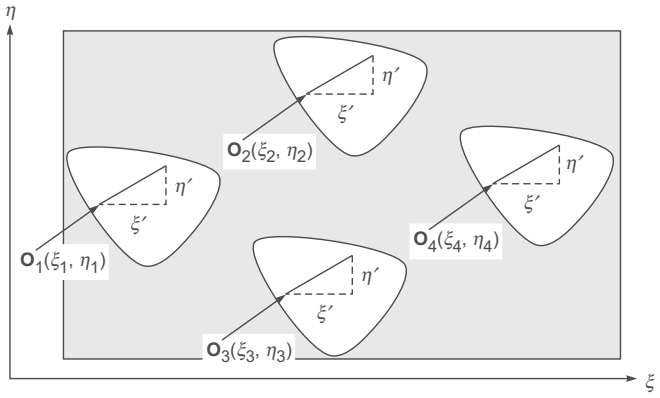


Fig. 19.7 Diffraction of a plane wave incident normally on an array of N identical apertures.

where each integral represents the contribution from a particular aperture. Let $O_1, O_2, O_3, \ldots$ represent points that are identically situated inside the apertures. For example, if the apertures are rectangular in nature, then $O_1, O_2, O_3, \ldots$ could represent the centers of the (rectangular) aperture. Let (ξ_1, η_1), (ξ_2, η_2), (ξ_3, η_3), … represent the coordinates of the points $O_1, O_2, O_3, \ldots$ respectively; then

$$u = C\sum_{n=1}^{N}\iint e^{-i[u(\xi_n+\xi')+v(\eta_n+\eta')]}\, d\xi'\, d\eta' \qquad \textbf{(19.47)}$$

where (ξ', η') represent the coordinates of an arbitrary point in a given aperture with respect to the point (ξ_n, η_n) as shown in Fig. 19.7. Thus,

$$u = u_s\sum_{n=1}^{N} e^{-i[u\xi_n+v\eta_n]} \qquad \textbf{(19.48)}$$

where

$$u_s = C\iint e^{-i(u\xi'+v\eta')}d\xi'\, d\eta' \qquad \textbf{(19.49)}$$

is the field produced by a single aperture. Thus, the resultant intensity distribution would be given by

$$I = I_s\, I_1 \qquad \textbf{(19.50)}$$

where I_s represent the intensity produced by a single aperture and

$$I_1 = \left|\sum_{n=1}^{N} e^{-i[u\xi_n+v\eta_n]}\right|^2 \qquad \textbf{(19.51)}$$

represents the intensity distribution produced by N point sources.

As an example, we consider N equally spaced identical apertures as shown in Fig. 19.8. Without loss of generality, we may assume

$$\xi_n = (n-1)d \quad \text{and} \quad \eta_n = 0; \quad n = 1, 2, 3 \ldots N$$

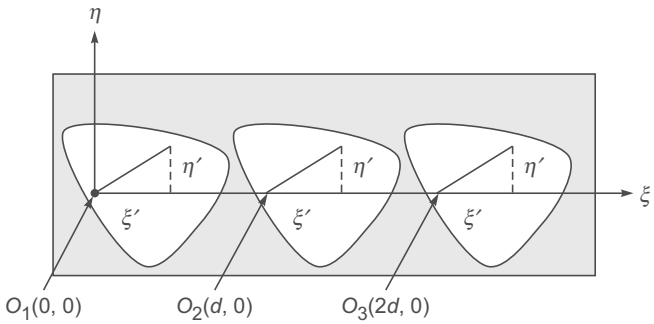


Fig. 19.8 Diffraction of a plane wave incident normally on an array of N identical equally spaced apertures.

Thus,

$$\sum_{n=1,2,3}^{N} e^{-iu(n-1)d} = 1 + e^{-iud} + \ldots + e^{-i(N-1)ud} = \frac{1-e^{-iNud}}{1-e^{-iud}}$$

$$= \exp\left[-\frac{1}{2}i(N-1)ud\right]\frac{\sin N\gamma}{\sin\gamma} \qquad \textbf{(19.52)}$$

where

$$\gamma = \frac{ud}{2} = \frac{\pi d\sin\theta}{\gamma} \qquad \textbf{(19.53)}$$

and

$$\sin\theta = \frac{x}{z} \qquad \textbf{(19.54)}$$

We therefore obtain

$$I_1 = \left| \sum_{n=1,2,3\ldots}^{N} e^{-iu(n-1)d} \right|^2 = \frac{\sin^2 N\gamma}{\sin^2 \gamma} \qquad \textbf{(19.55)}$$

which is the interference pattern produced by N identically placed point sources; this is the same result as derived in Sec. 18.7. When $N = 2$, we obtain the interference pattern produced by 2 point sources.

If each aperture is a long narrow slit, we obtain the diffraction pattern produced by a grating [see Eq. (18.50)]. On the other hand, if each aperture is circular we obtain the product of the Airy pattern and the interference pattern produced by two point sources [see Figs. 17.4 and 17.5].

19.9 SPATIAL FREQUENCY FILTERING

LO 5

In the next section, we will show that if $g(x, y)$ represents the field distribution on the front focal plane of a corrected lens (i.e., on the plane P_1 in Fig. 19.9) then on the back focal plane P_2 of the lens, one obtains the Fourier transform of $g(x, y)$, the z-axis represents the optical axis of the lens. Thus if $G(x, y)$ represents the field distribution on the back focal plane P_2 then it is related to $g(x, y)$ through the following relation:

$$G(u, v) = \frac{1}{\lambda f} \iint g(x', y') \exp[-i(ux' + vy')]\, dx'\, dy' \qquad \textbf{(19.56)}$$

where

$$u \equiv \frac{2\pi x}{\lambda f} \quad \text{and} \quad v \equiv \frac{2\pi y}{\lambda f} \qquad \textbf{(19.57)}$$

represent the spatial frequencies. Further, λ represents the wavelength of light and f is the focal length of the lens. If we compare Eq. (19.56) with Eq. (9.39) of Chapter 9, we find that

the field distribution on the back focal plane of a corrected lens is the Fourier transform of the field distribution on the front plane.

This important property of a corrected lens forms the basis of the subject of spatial frequency filtering which finds applications in many diverse areas (see, e.g., Ref. 19.2, 19.4, 19.8–19.10). We must mention here that in writing Eq. (19.56), we have neglected an (unimportant) phase factor on the right-hand side [see Sec. 19.11].

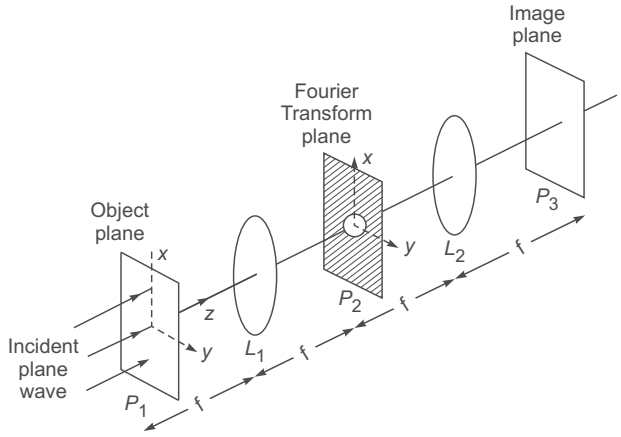


Fig. 19.9 The plane P_2 is the Fourier transform plane where the spatial frequency components of the object (placed in the plane P_1) are displayed. In the above figure, a small hole is placed on the axis (in the plane P_2) which filters out the high-frequency components.

We first consider a plane wave incident normally on the lens. This implies that $g(x, y)$ is a constant (= g_0, say) and

$$G(u, v) = \frac{g_0}{\lambda f} \iint \exp[-i(ux' + vy')]\, dx'\, dy' \qquad \textbf{(19.58)}$$

Now, if we use

$$\int_{-\infty}^{+\infty} e^{-iux}\, dx = 2\pi\, \delta(u) \qquad \textbf{(19.59)}$$

[see Eq. (9.9) of Chapter 9], we would obtain

$$G(u, v) = \frac{g_0}{\lambda f} 4\pi^2\, \delta(u)\, \delta(v) \qquad \textbf{(19.60)}$$

where $\delta(u)$ and $\delta(v)$ represent the Dirac delta functions. Since $\delta(u) = 0$ for $u \neq 0$, one can infer from Eq. (19.60) that the intensity is zero at all points excepting at the point $x = 0$, $y = 0$. This is to be expected because a plane wave gets focused to a point by a corrected lens*.

Another interesting example is a one-dimensional cosinusoidal field distribution in the object plane i.e.,

$$g(x, y) = g_0 \cos(2\pi\alpha x) \qquad \textbf{(19.61)}$$

where α is a constant**; we have assumed no y-dependence of the field. If we use the identity

$$\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$$

* We are assuming a very large dimension of the aperture of the lens; as such the limits of integration in Eq. (19.56) are assumed to be from $-\infty$ to $+\infty$. This is a good approximation in most cases.

** On the plane P_1 (see Fig. 19.7), if we place the negative of the photograph shown in Fig. 14.11(b) with the y-axis along the length of a fringe and assume a plane wave to be incident normally on the film, then the field distribution would be proportional to $\cos^2(2\pi\alpha x)$ which is equal to $\frac{1}{2}[1 + \cos(2\pi\alpha x)]$.